

REPORTE DE EJECUCIÓN
ALGORITMO GENÉTICO EN ESPACIO 2D Y 3D

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MÉTODOS UTILIZADOS:

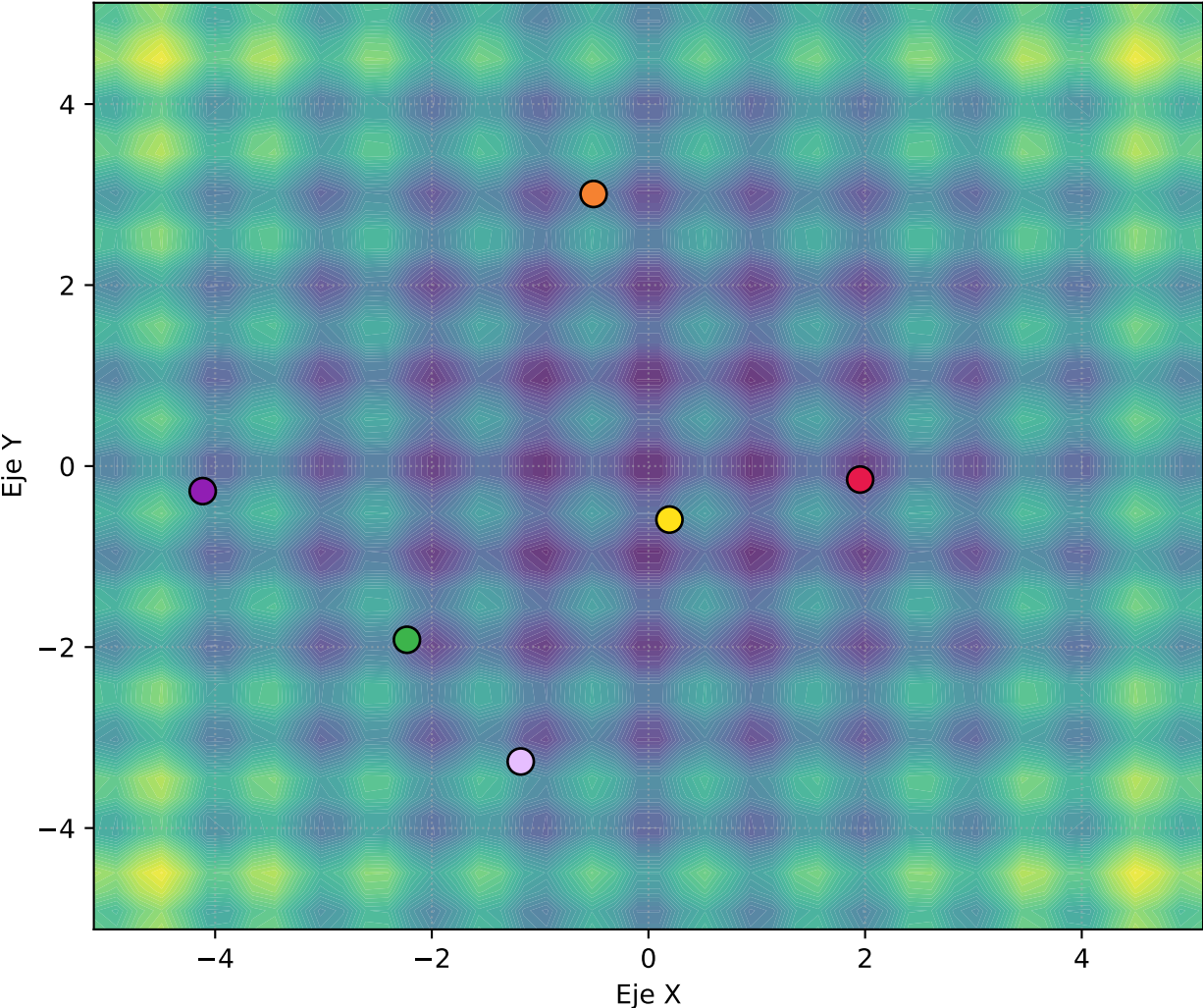
- > Aptitud 3D: □ RASTRIGIN
- > Codificación: □ GRAY_2D
- > Selección: □ TORNEO
- > Cruza: □ UN_PUNTO
- > Mutación: □ UN_BIT
- > Generaciones: □ 100

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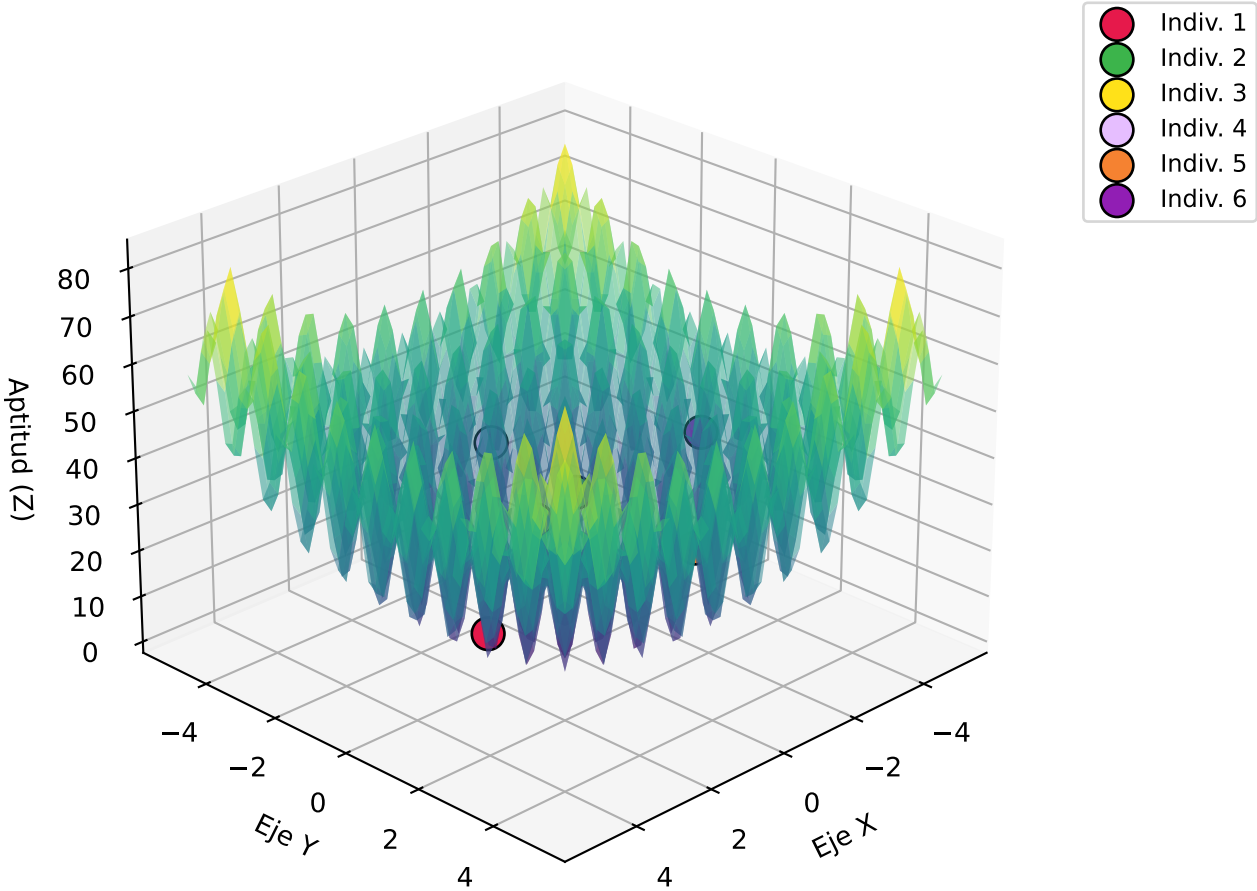
Centro de Investigaciones en Óptica

Generación 1: Posición Inicial de los 6 Mejores

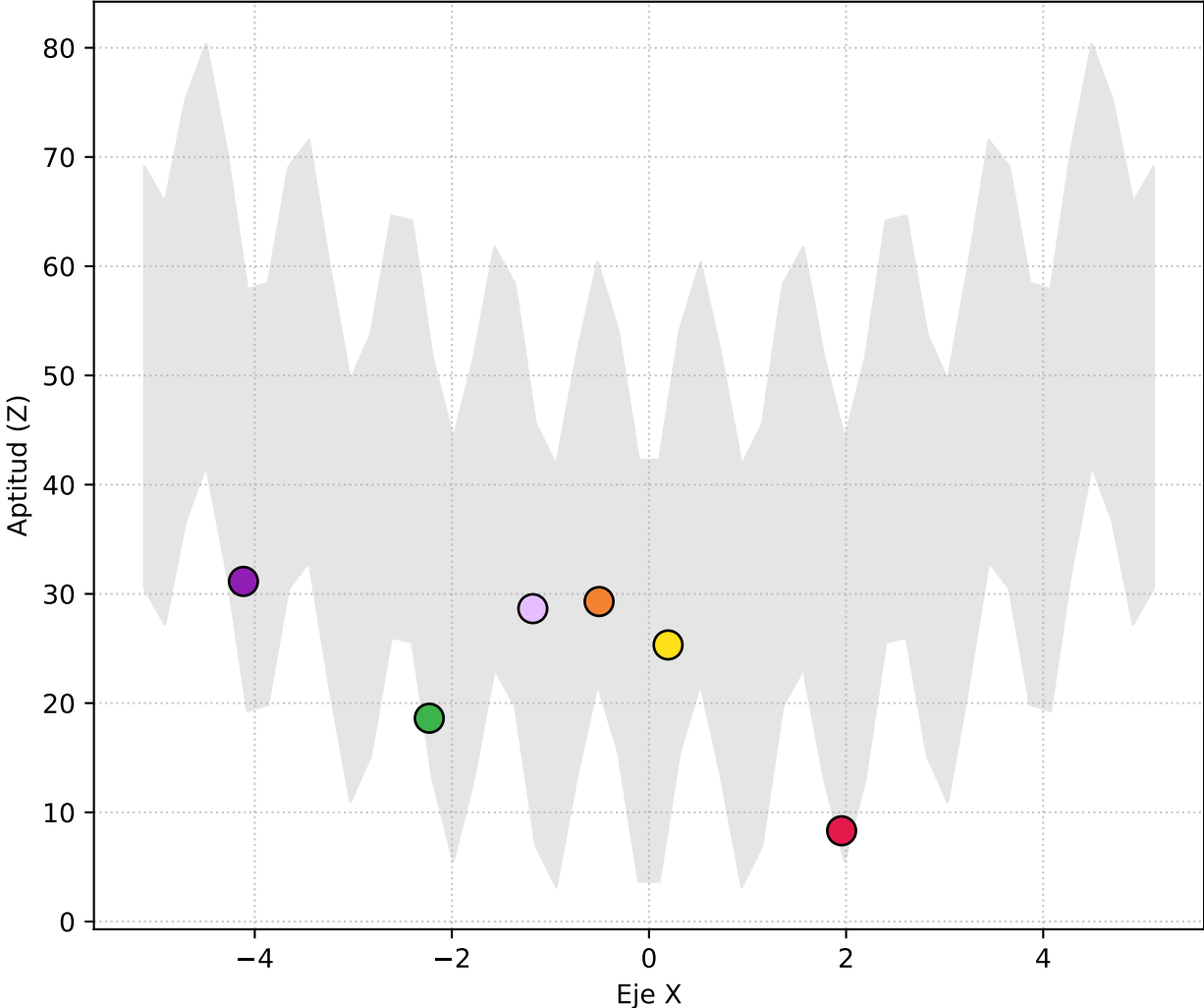
Vista Superior (X, Y)



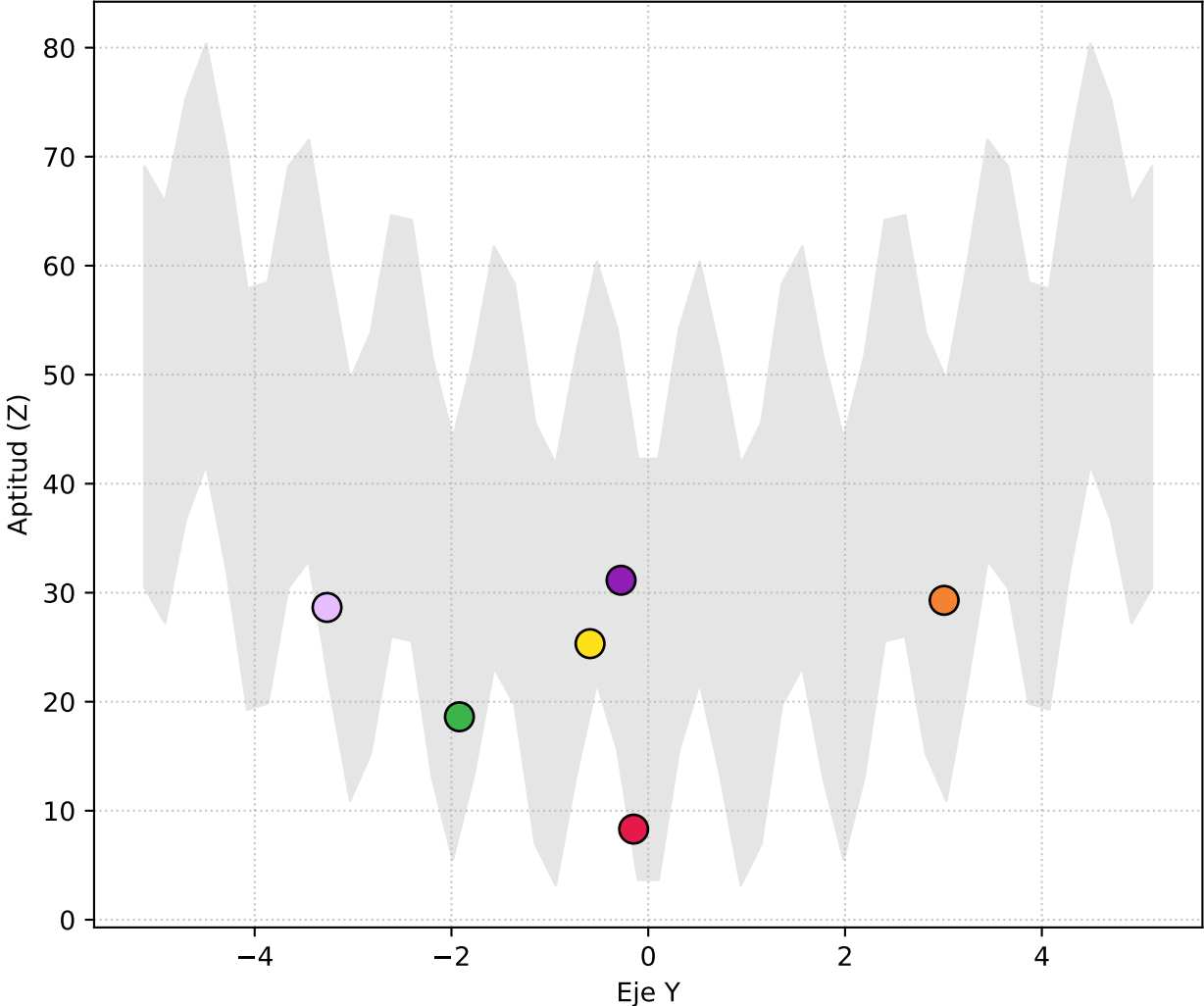
Perspectiva 3D



Vista Lateral (X vs Elevación Z)

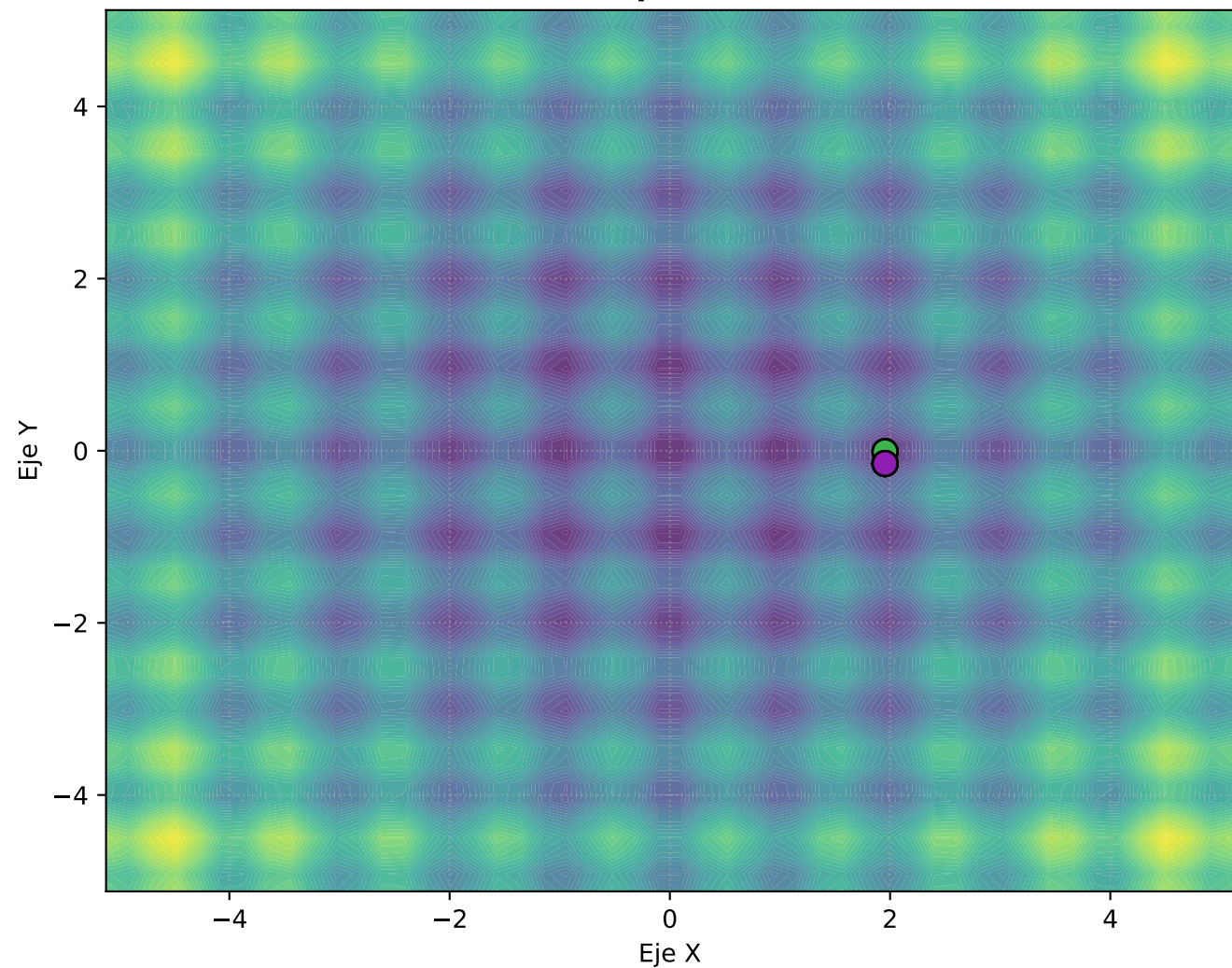


Vista Frontal (Y vs Elevación Z)

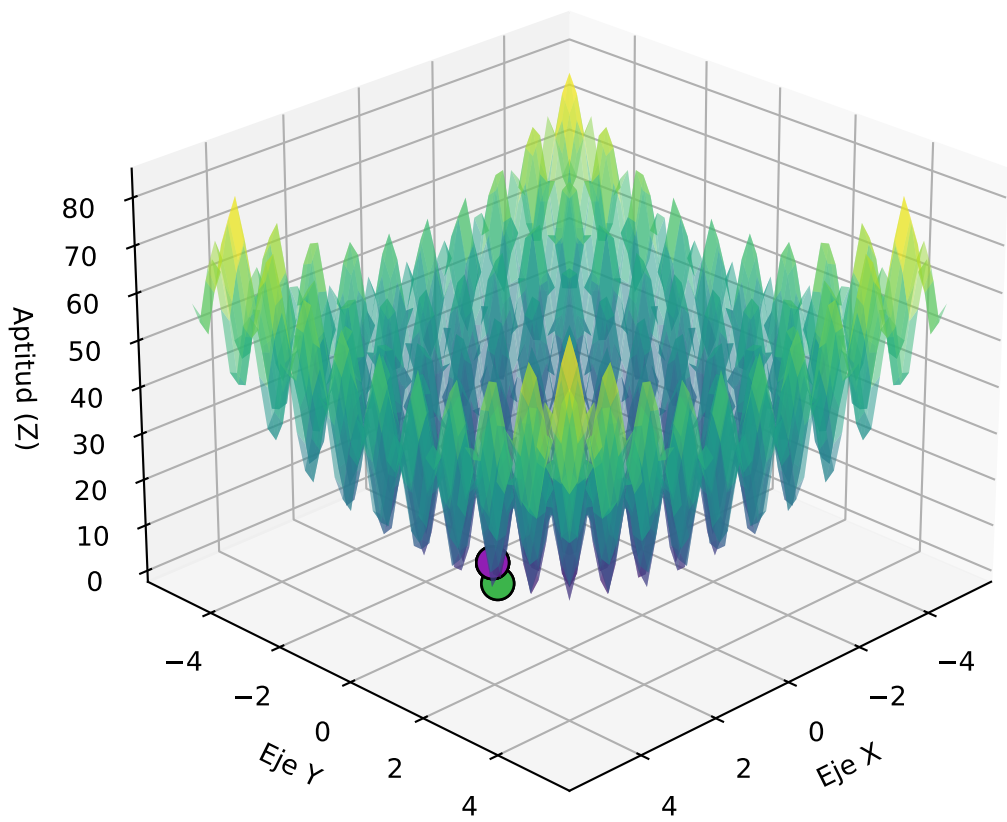


Progreso: Generación 10

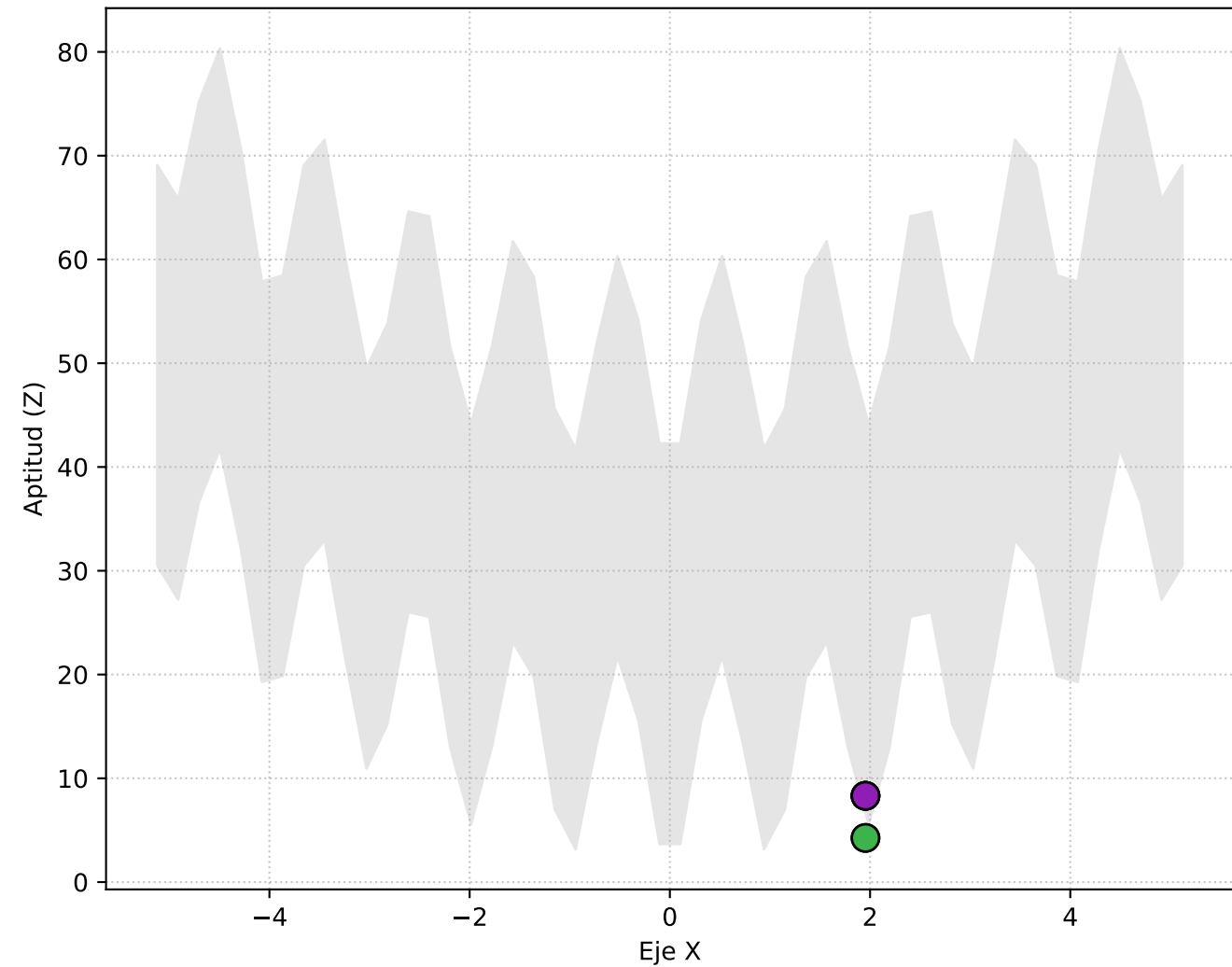
Vista Superior (X, Y)



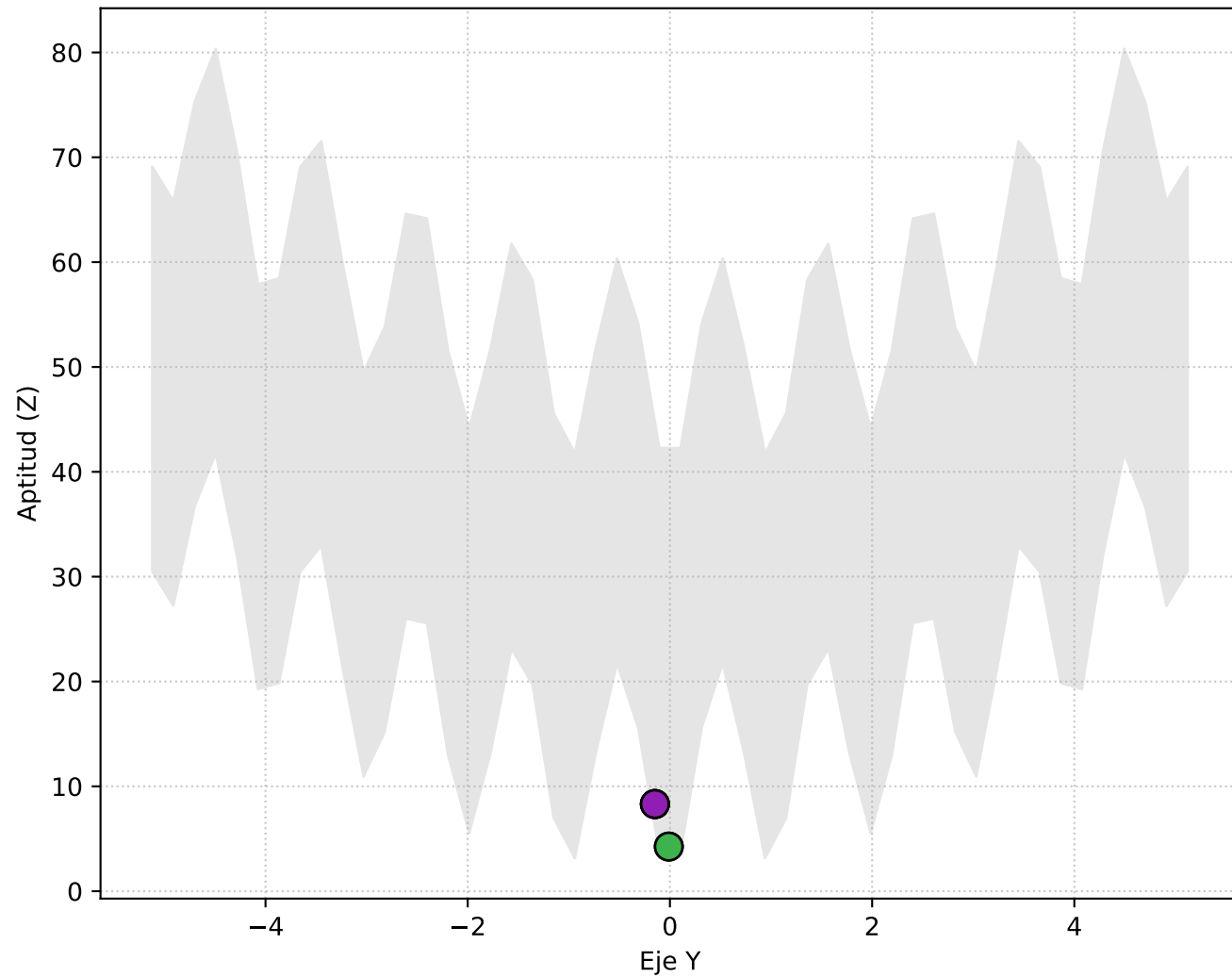
Perspectiva 3D



Vista Lateral (X vs Elevación Z)

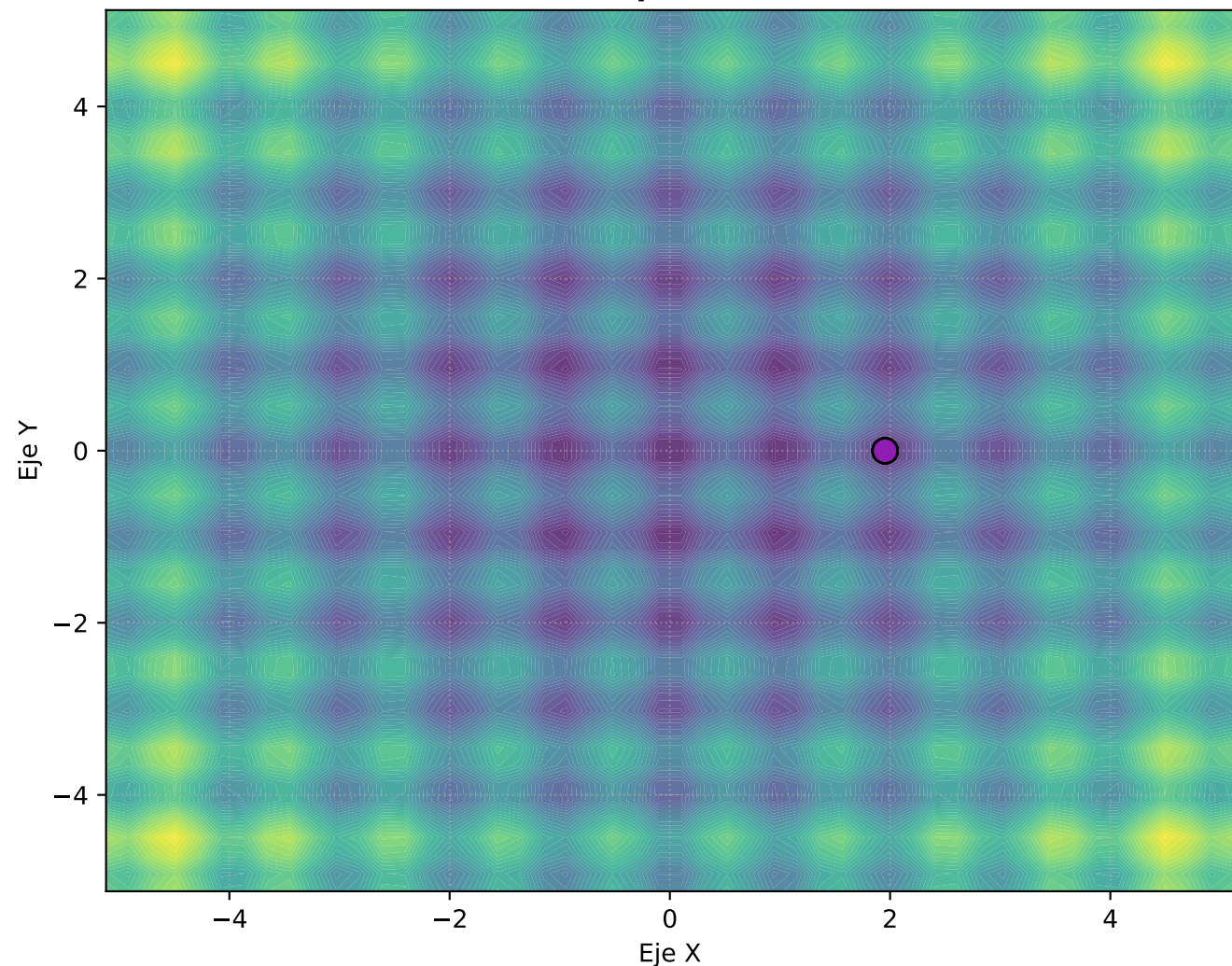


Vista Frontal (Y vs Elevación Z)

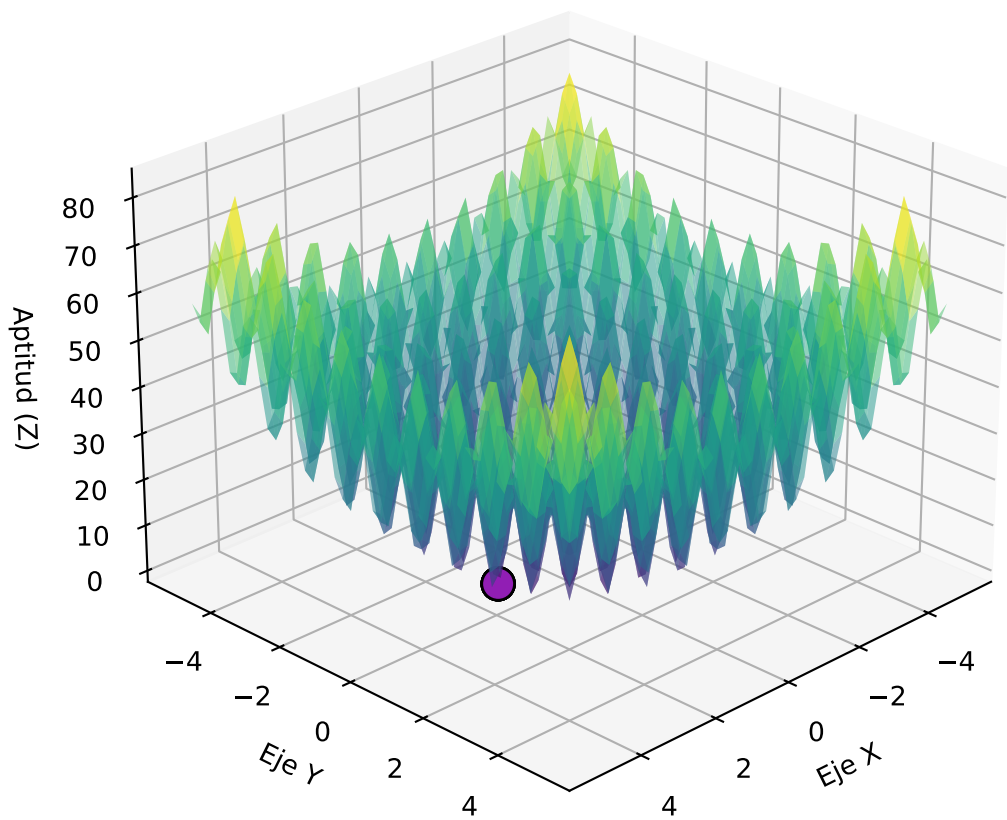


Progreso: Generación 20

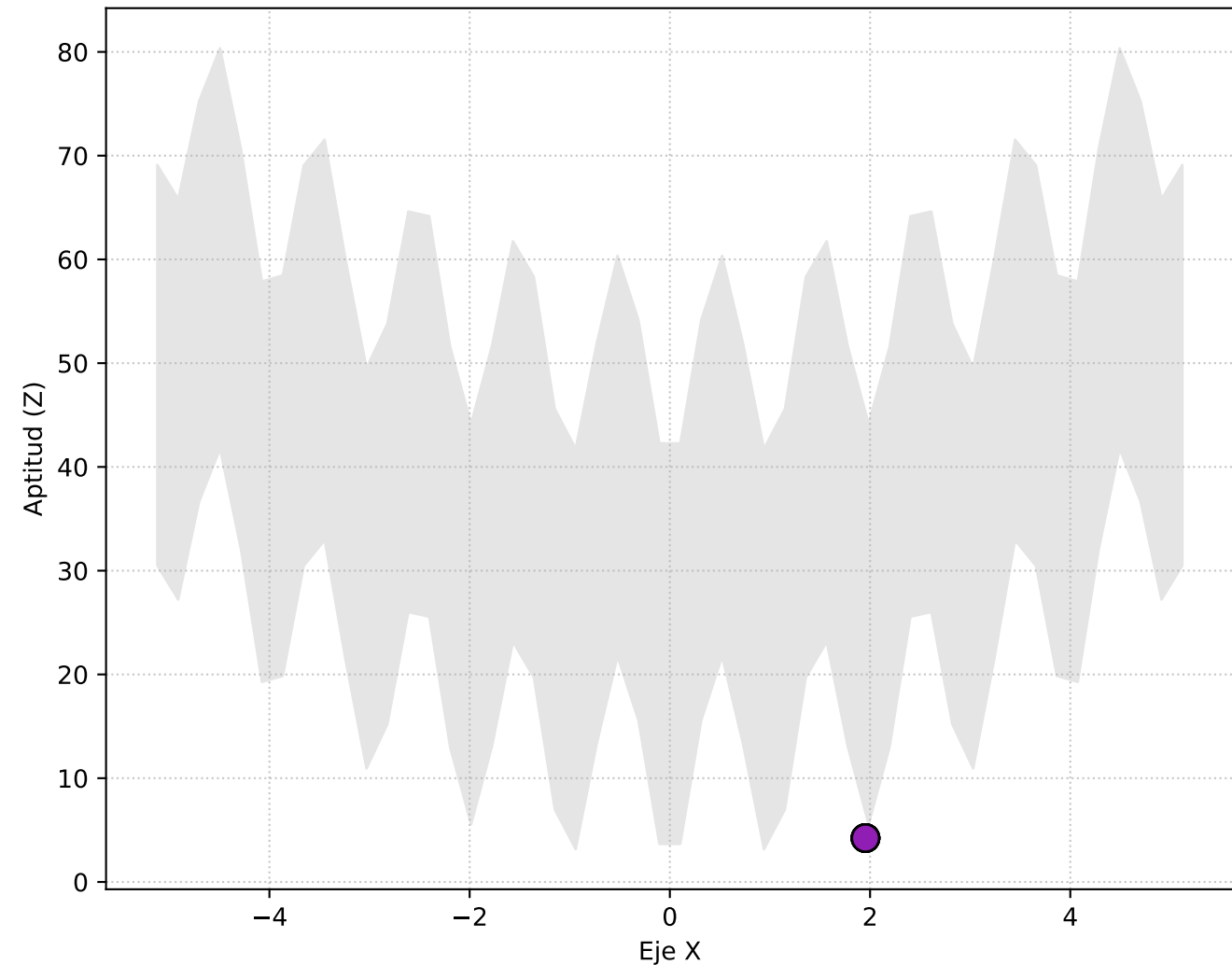
Vista Superior (X, Y)



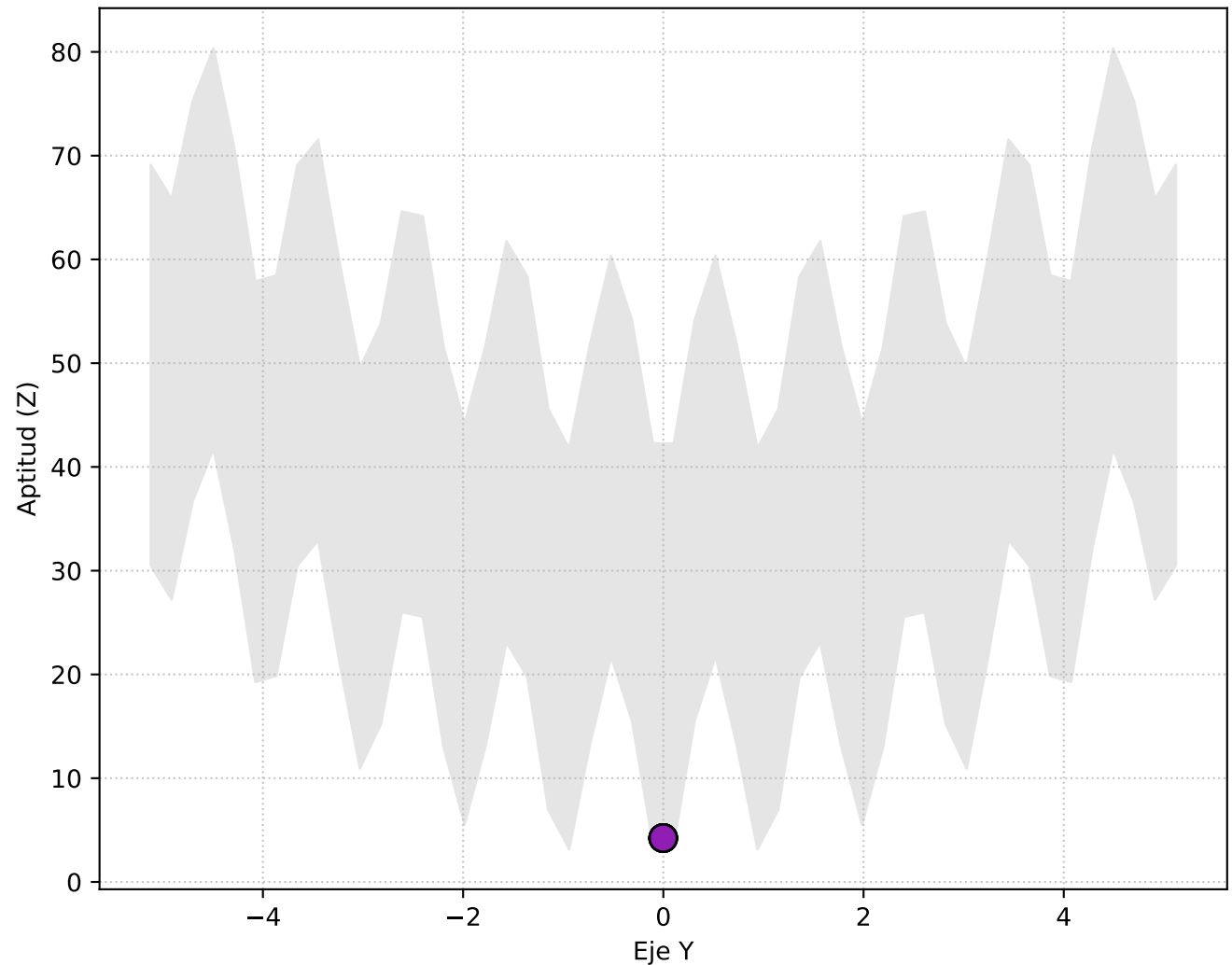
Perspectiva 3D



Vista Lateral (X vs Elevación Z)

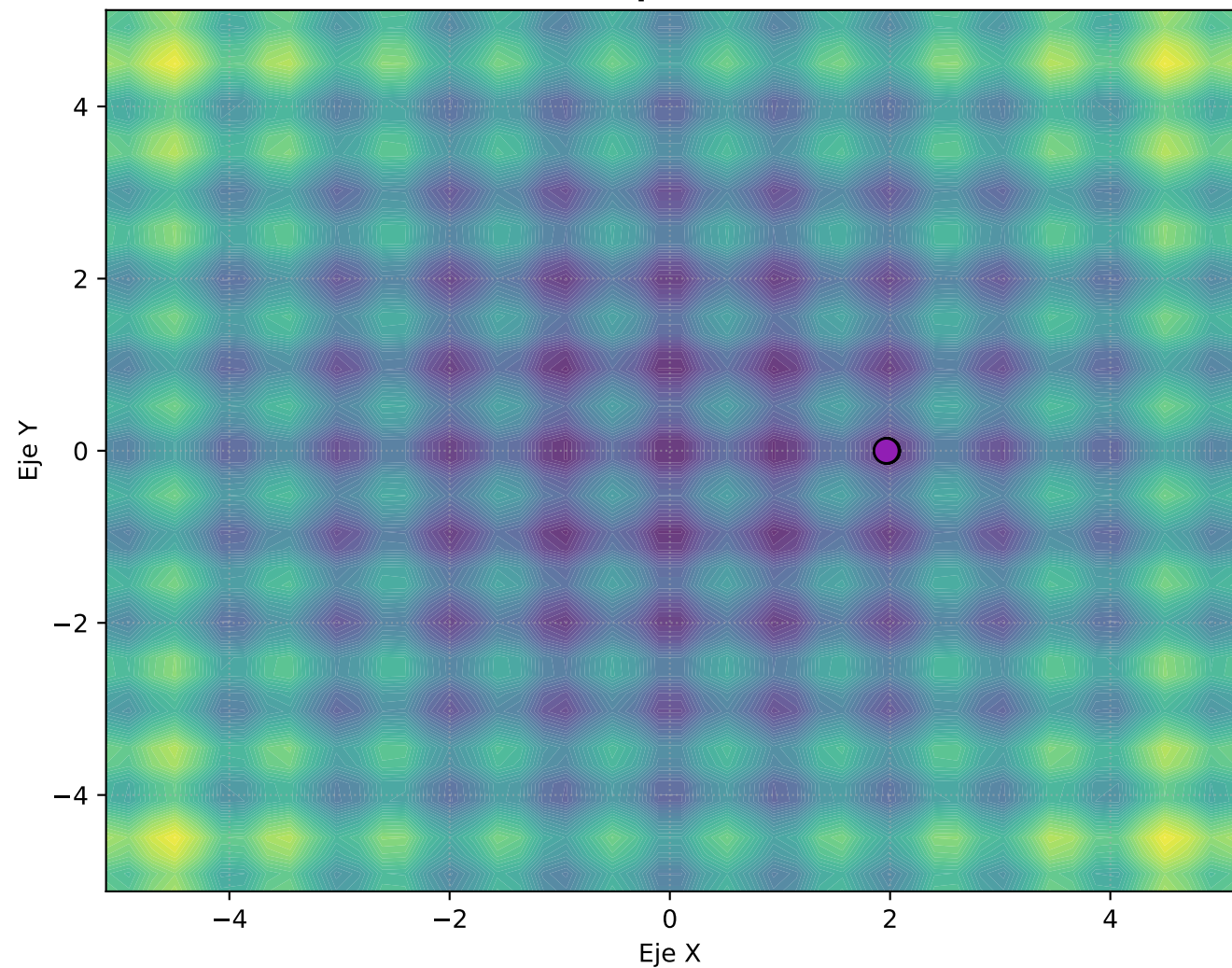


Vista Frontal (Y vs Elevación Z)

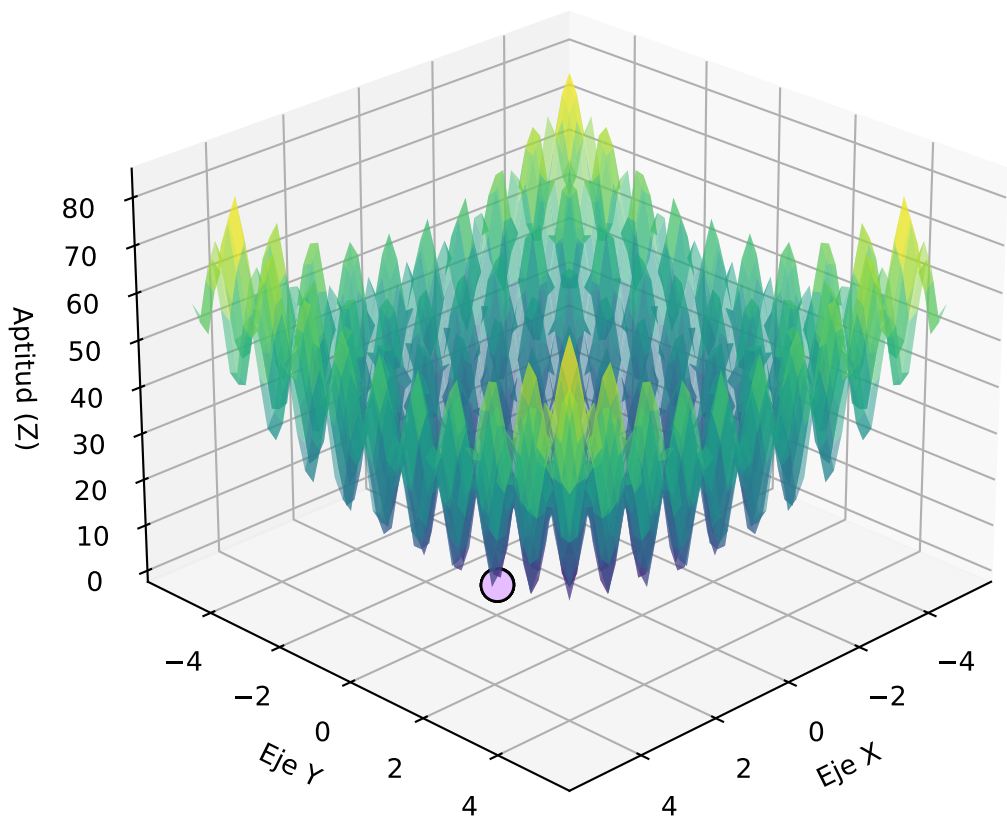


Progreso: Generación 30

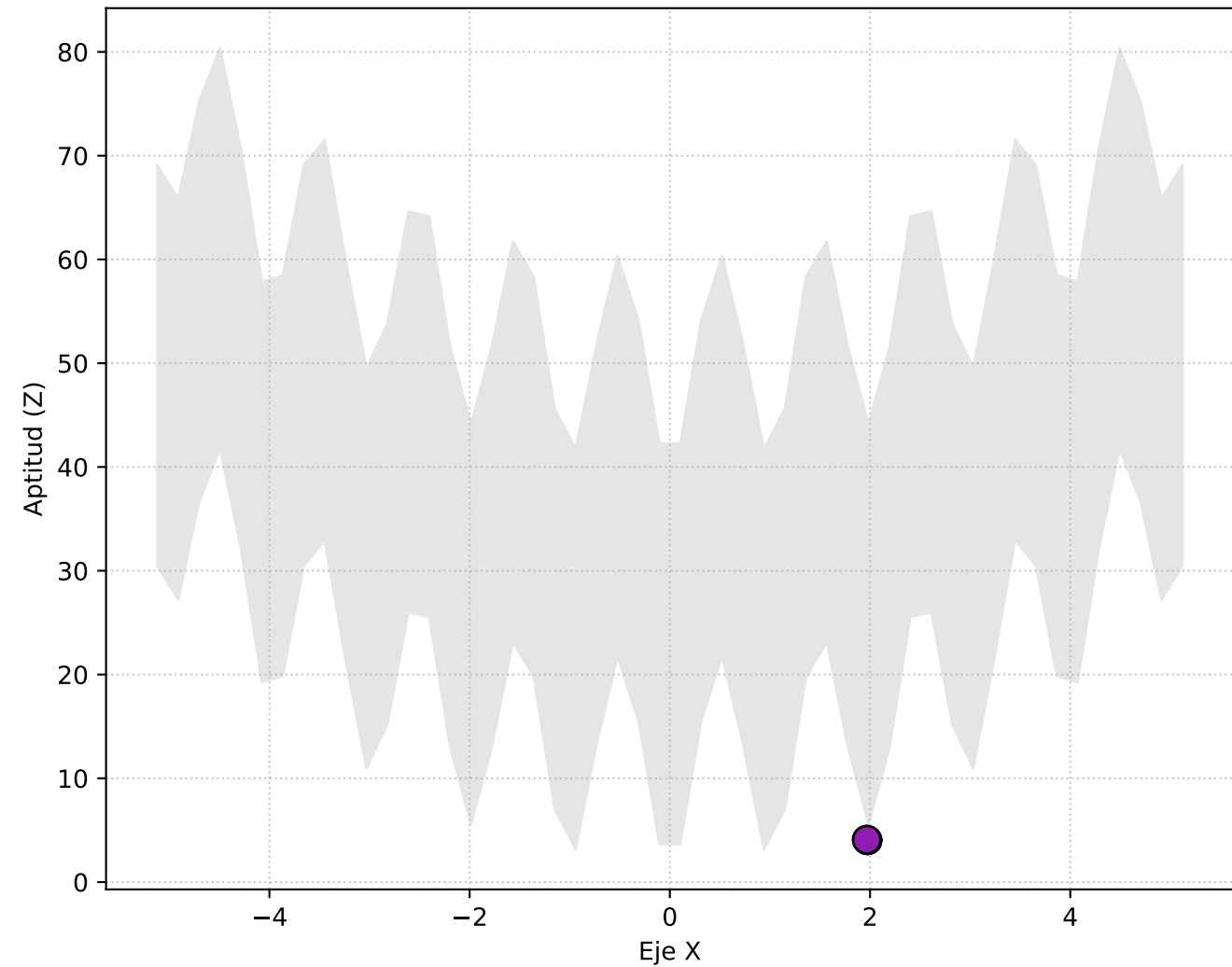
Vista Superior (X, Y)



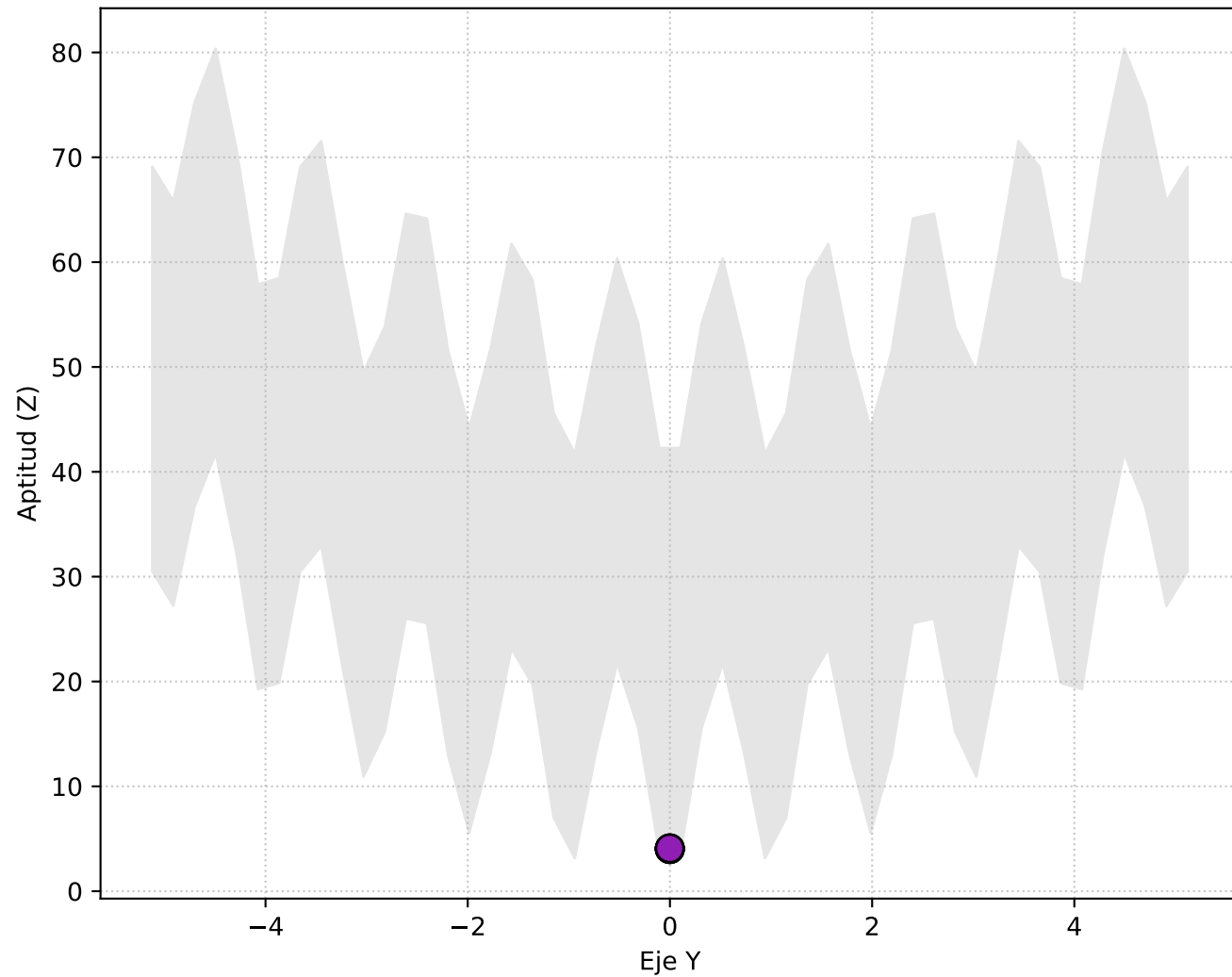
Perspectiva 3D



Vista Lateral (X vs Elevación Z)

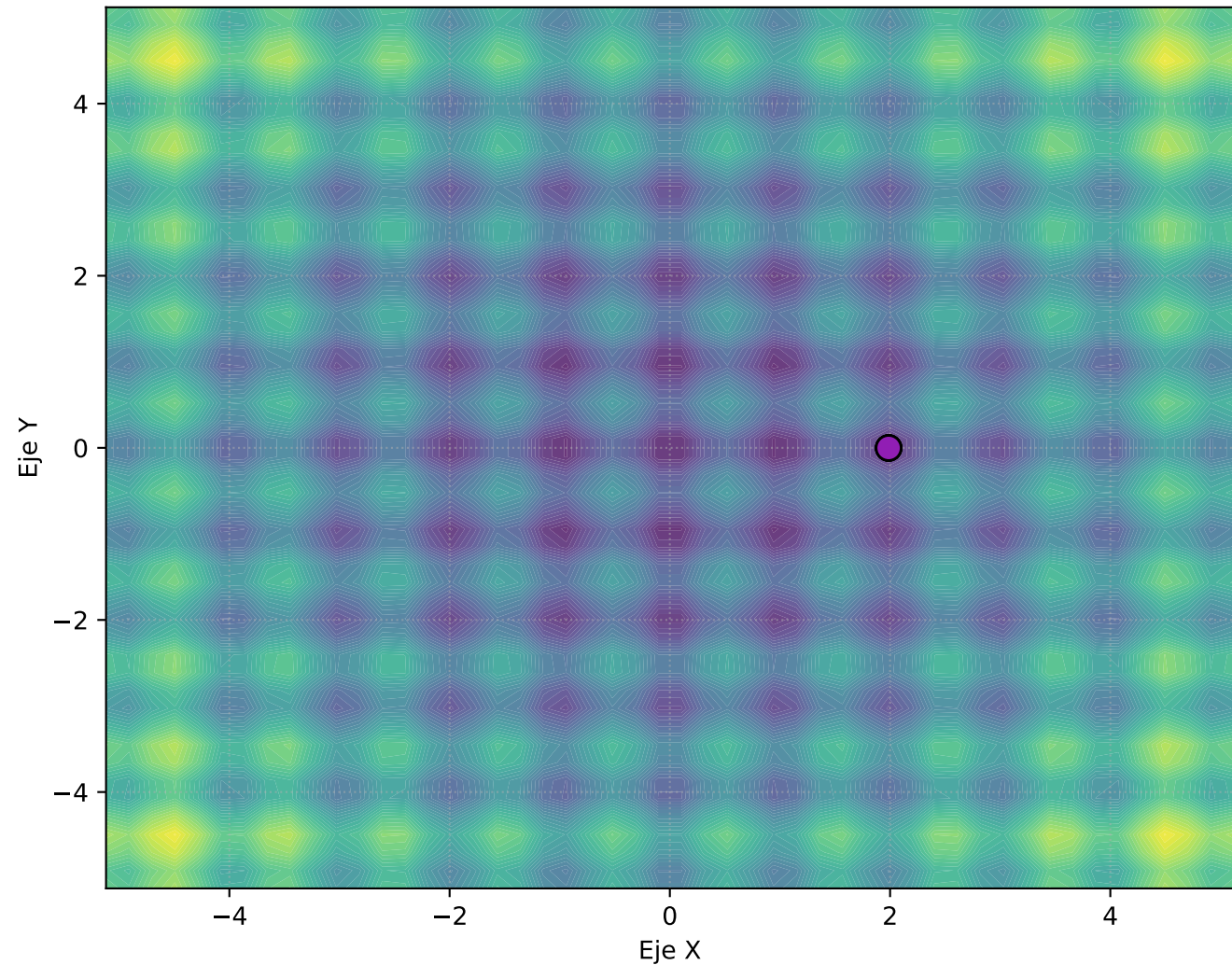


Vista Frontal (Y vs Elevación Z)

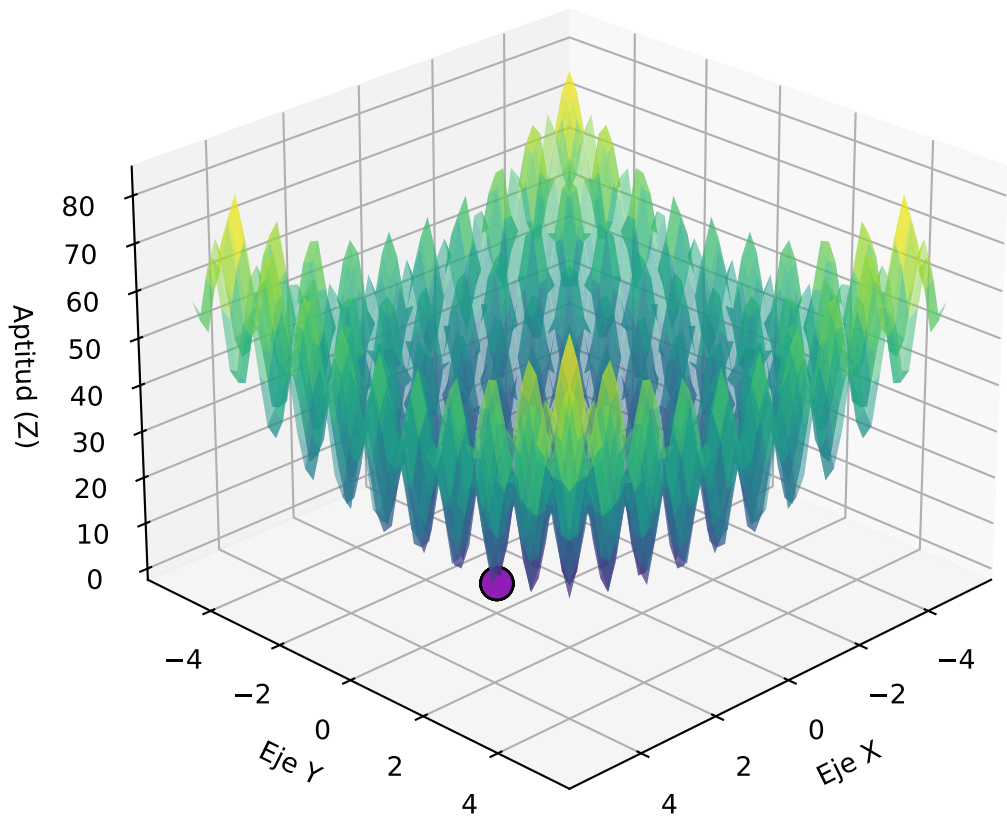


Progreso: Generación 40

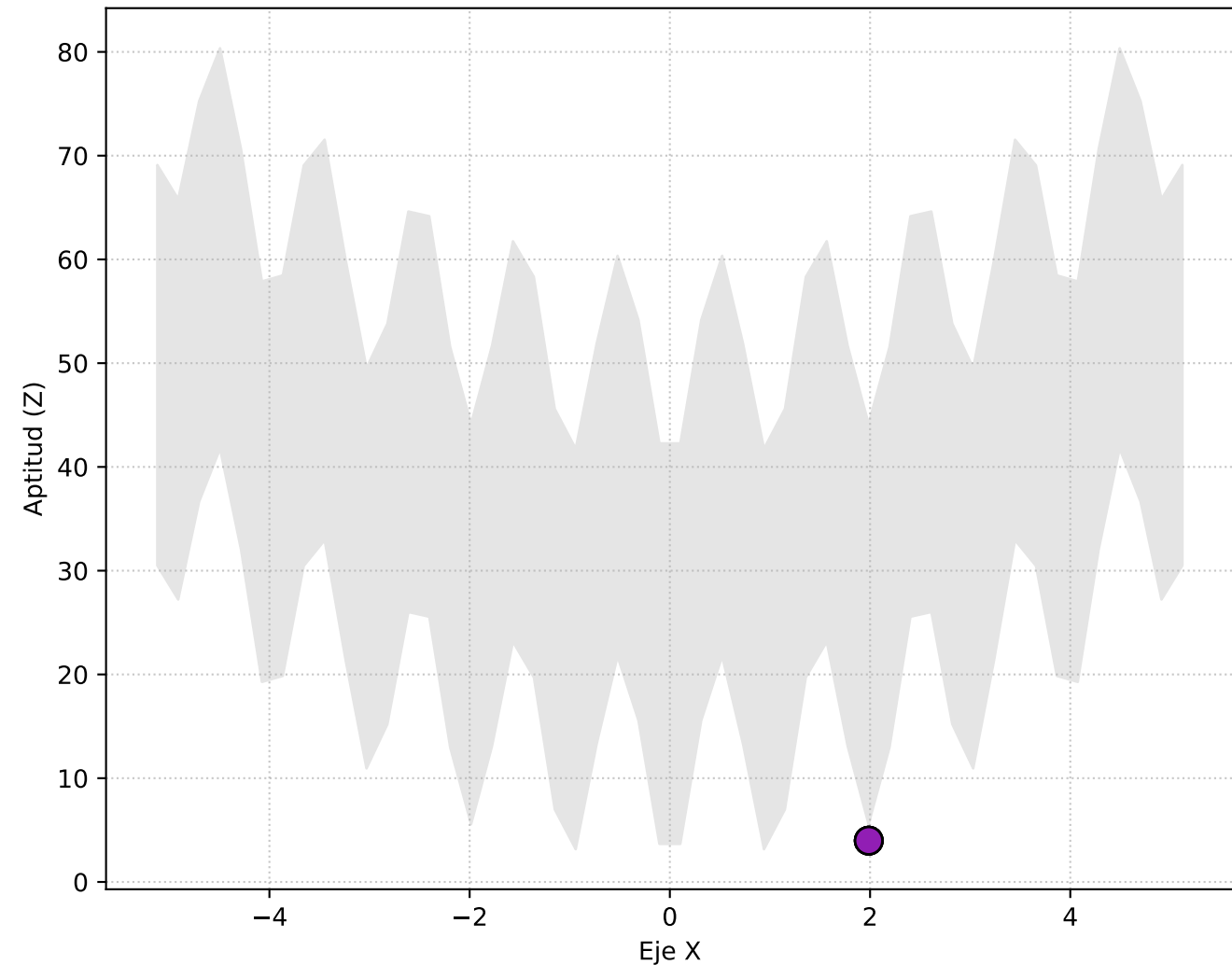
Vista Superior (X, Y)



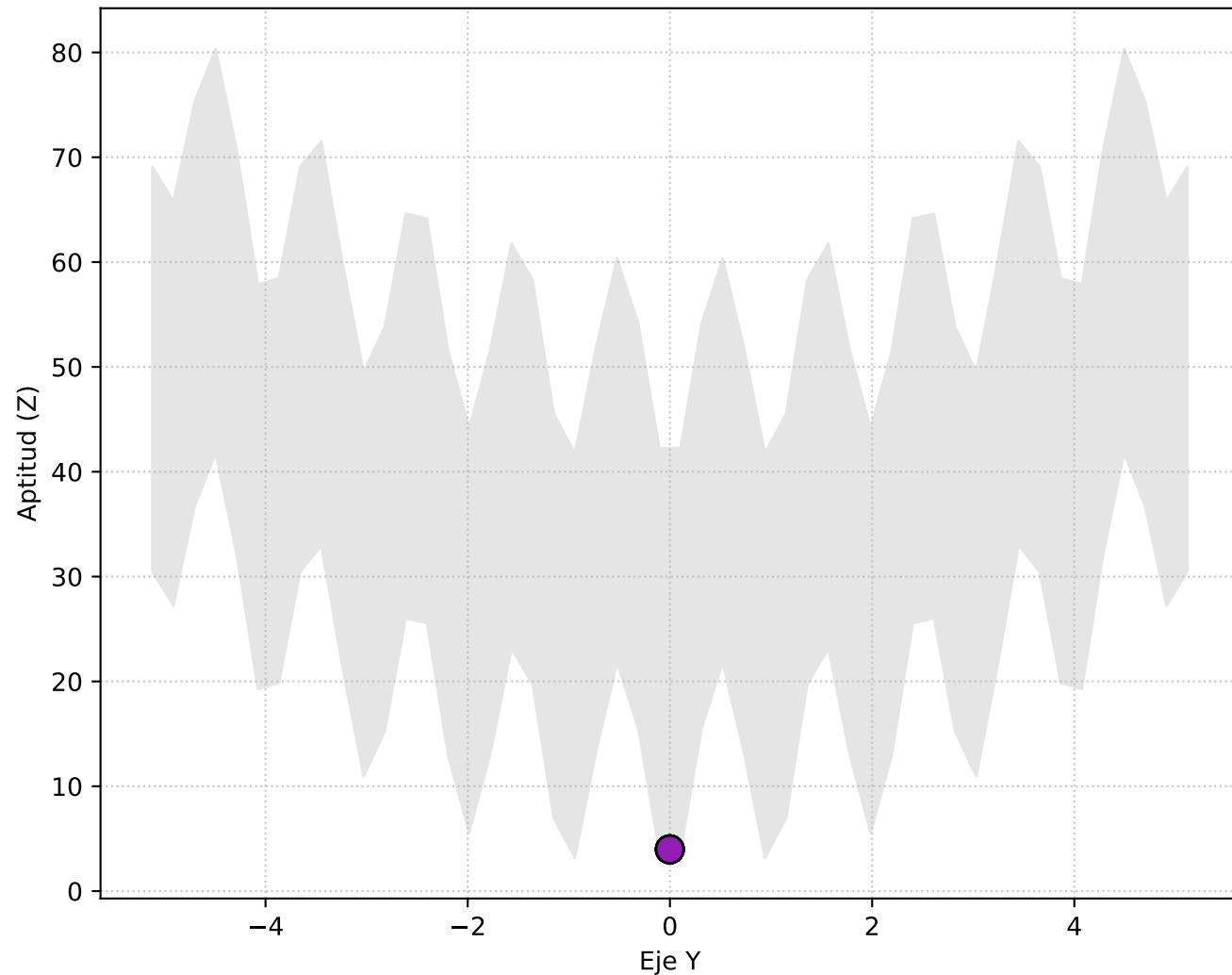
Perspectiva 3D



Vista Lateral (X vs Elevación Z)

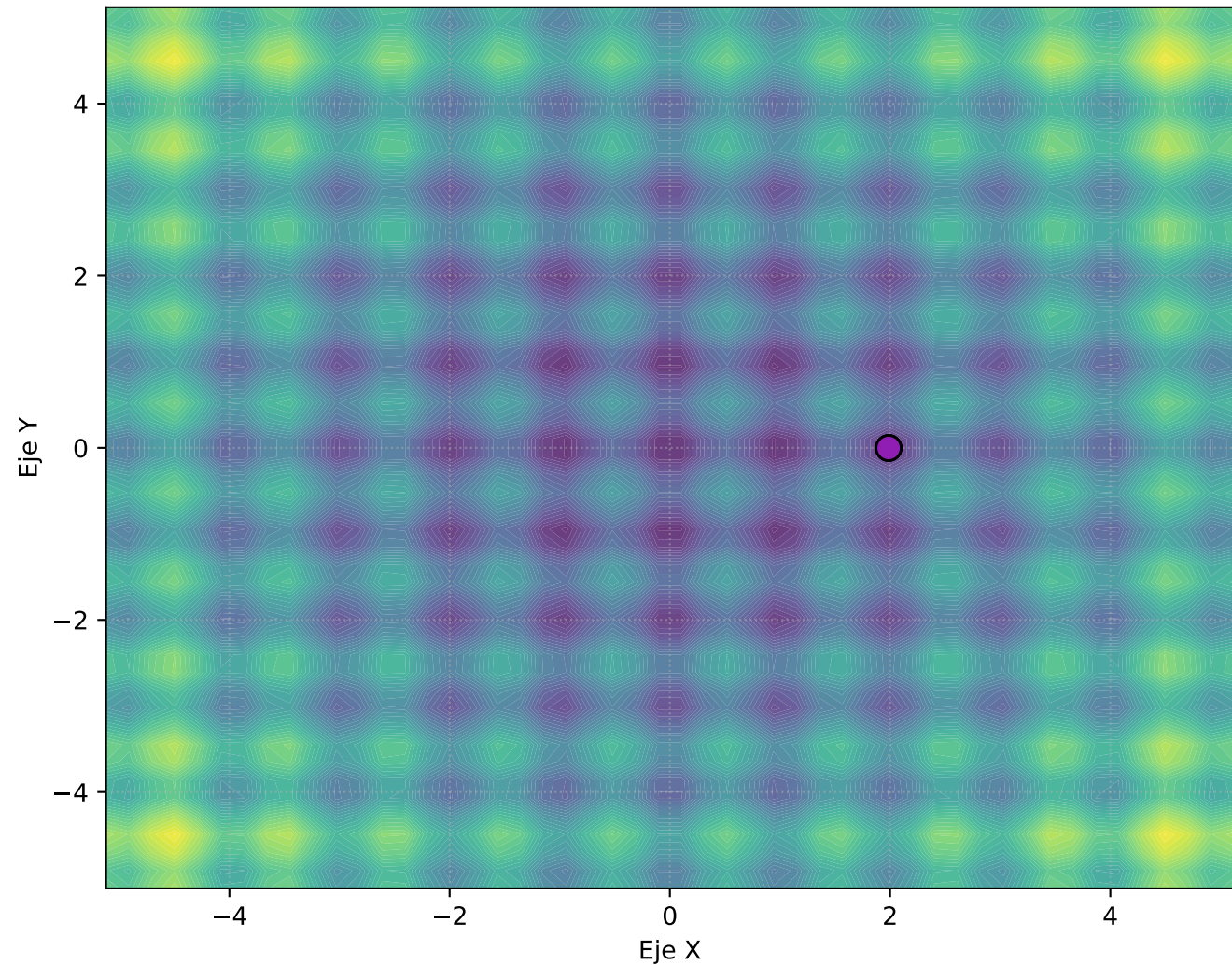


Vista Frontal (Y vs Elevación Z)

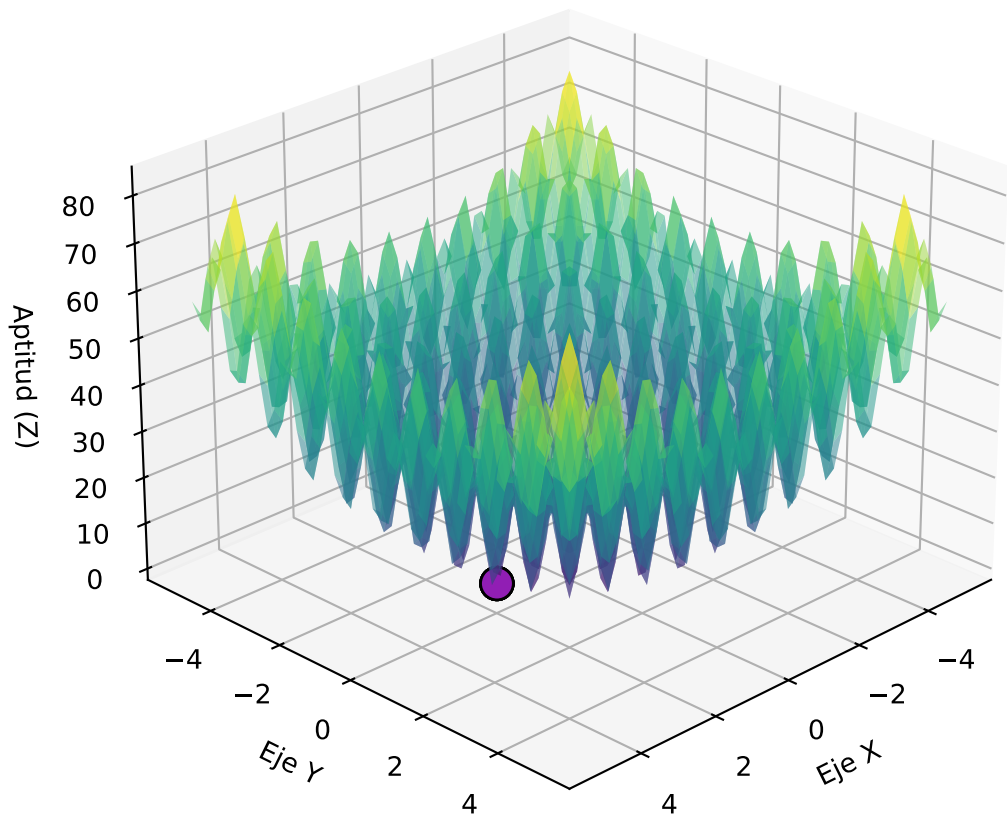


Progreso: Generación 50

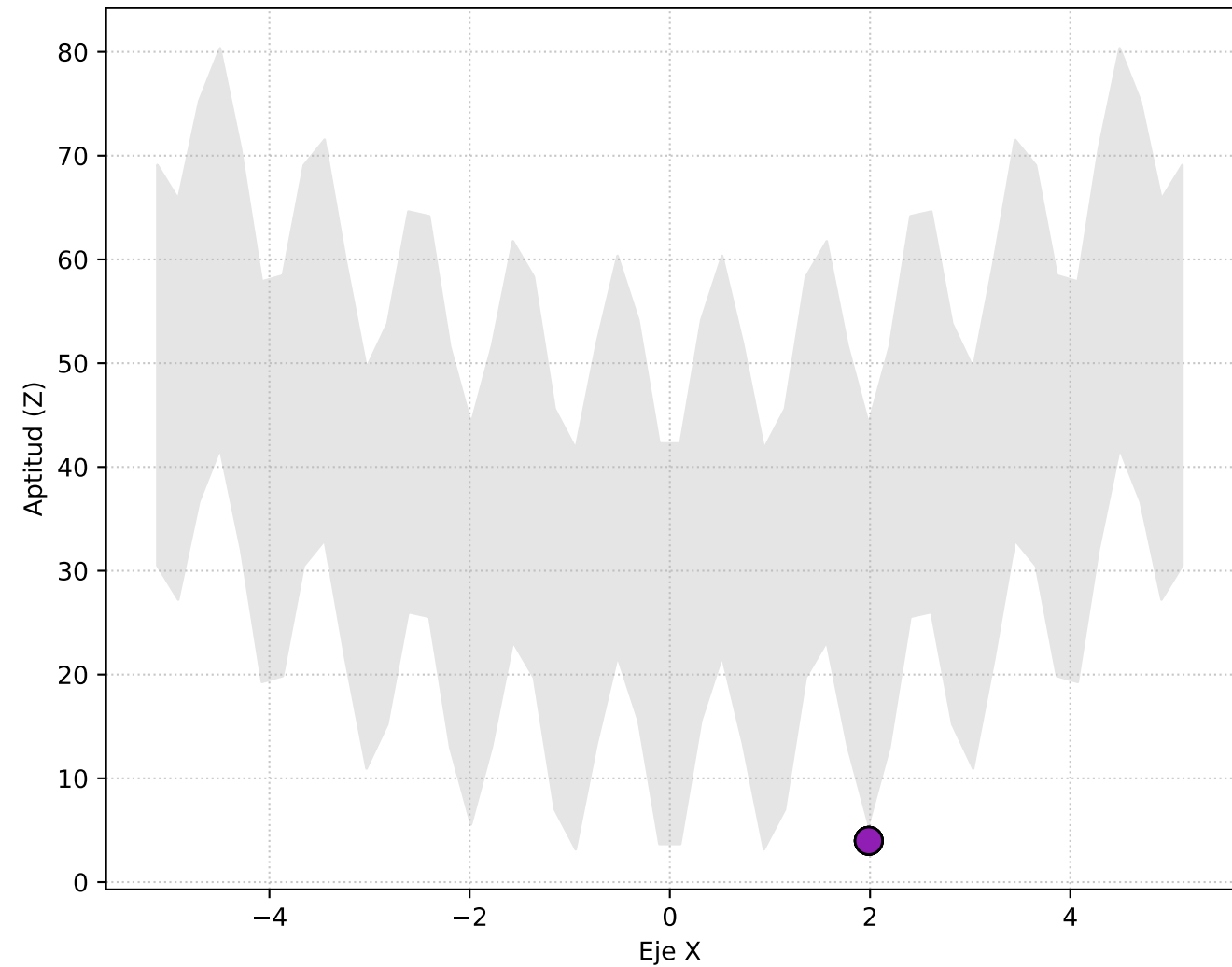
Vista Superior (X, Y)



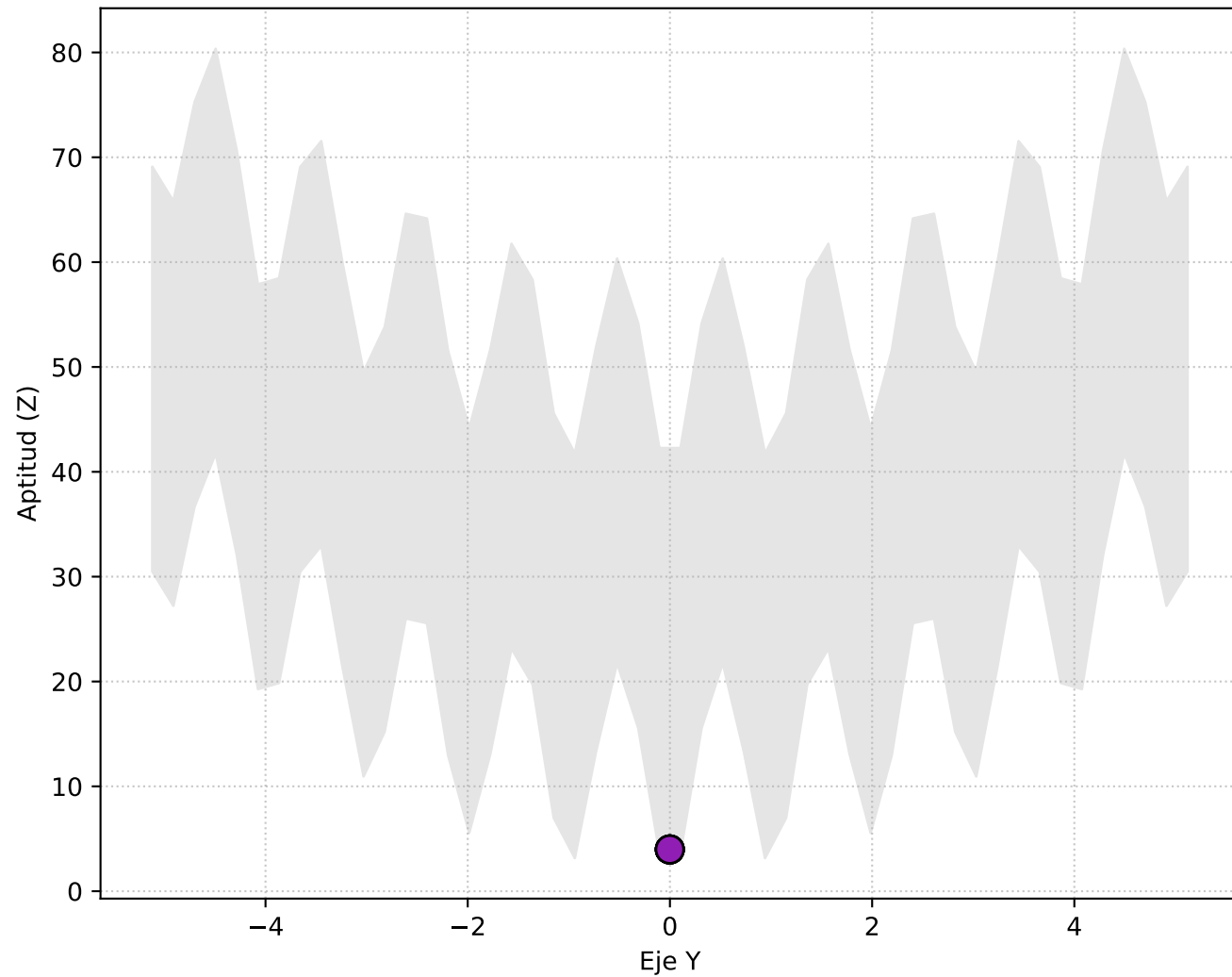
Perspectiva 3D



Vista Lateral (X vs Elevación Z)

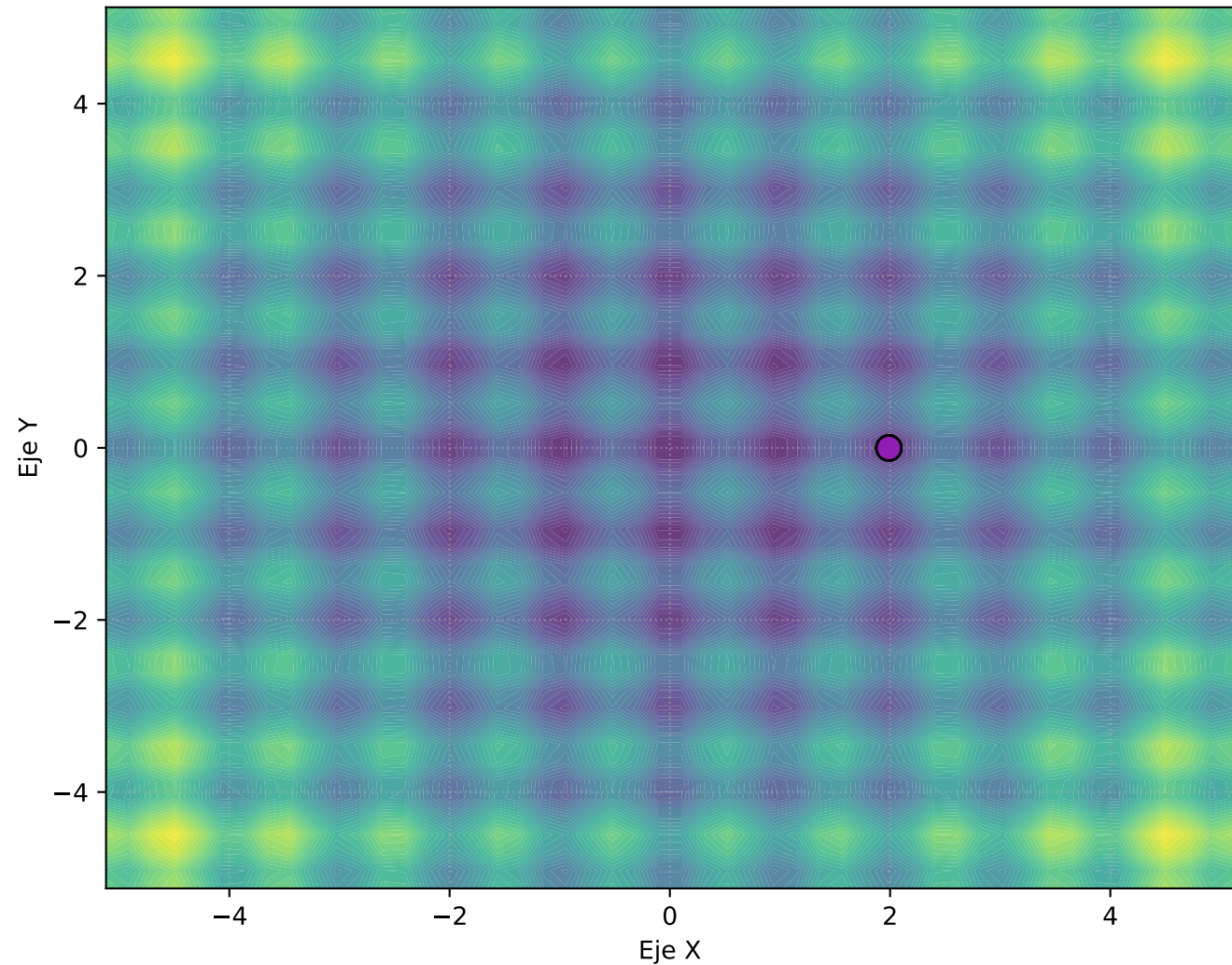


Vista Frontal (Y vs Elevación Z)

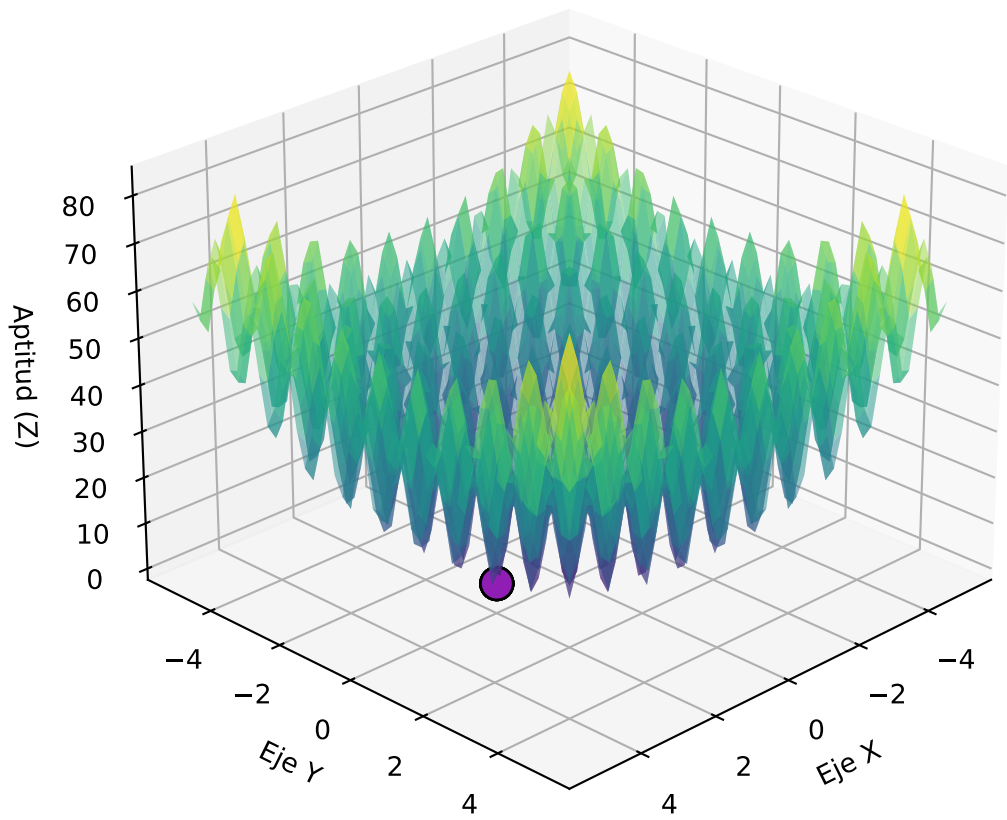


Progreso: Generación 60

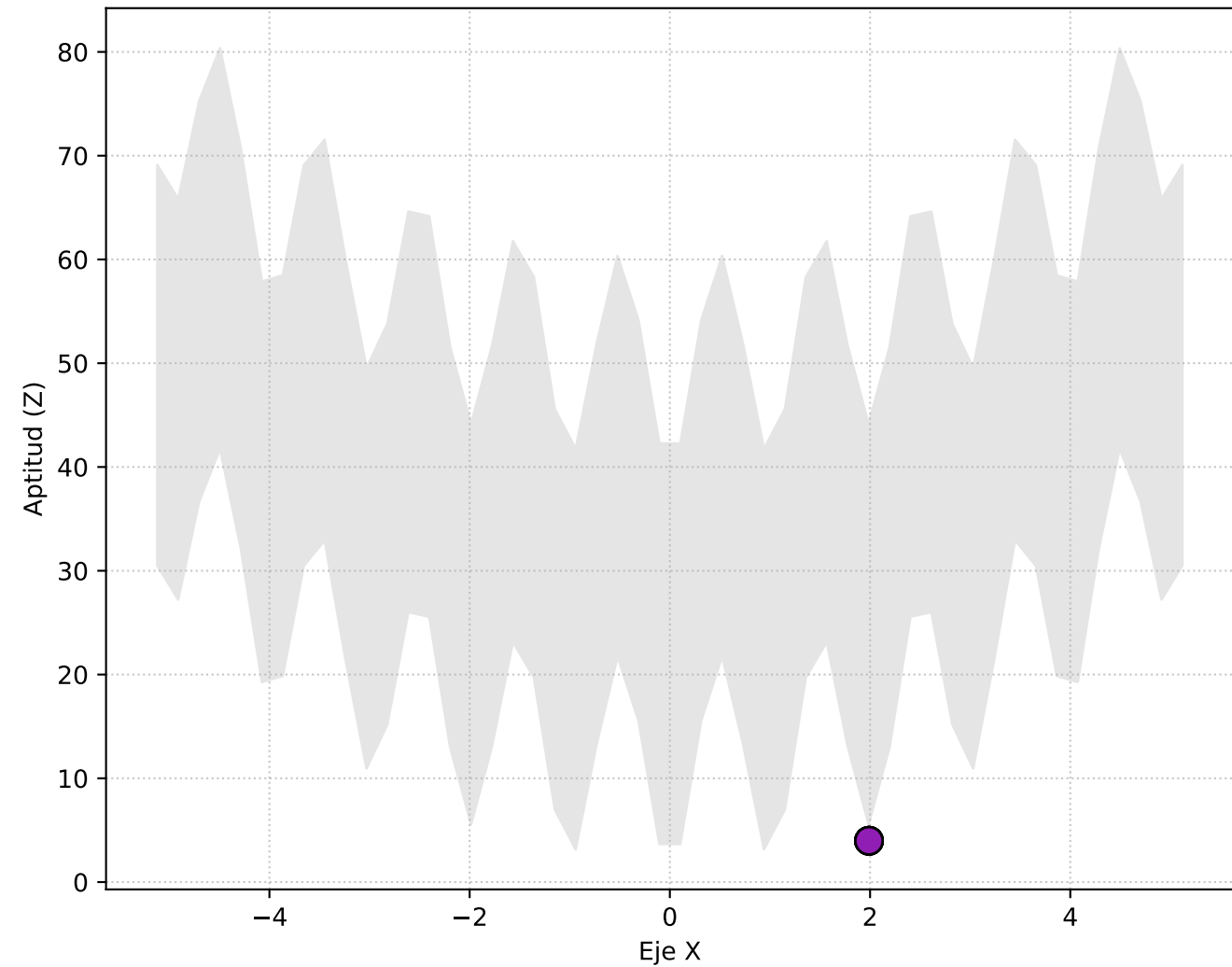
Vista Superior (X, Y)



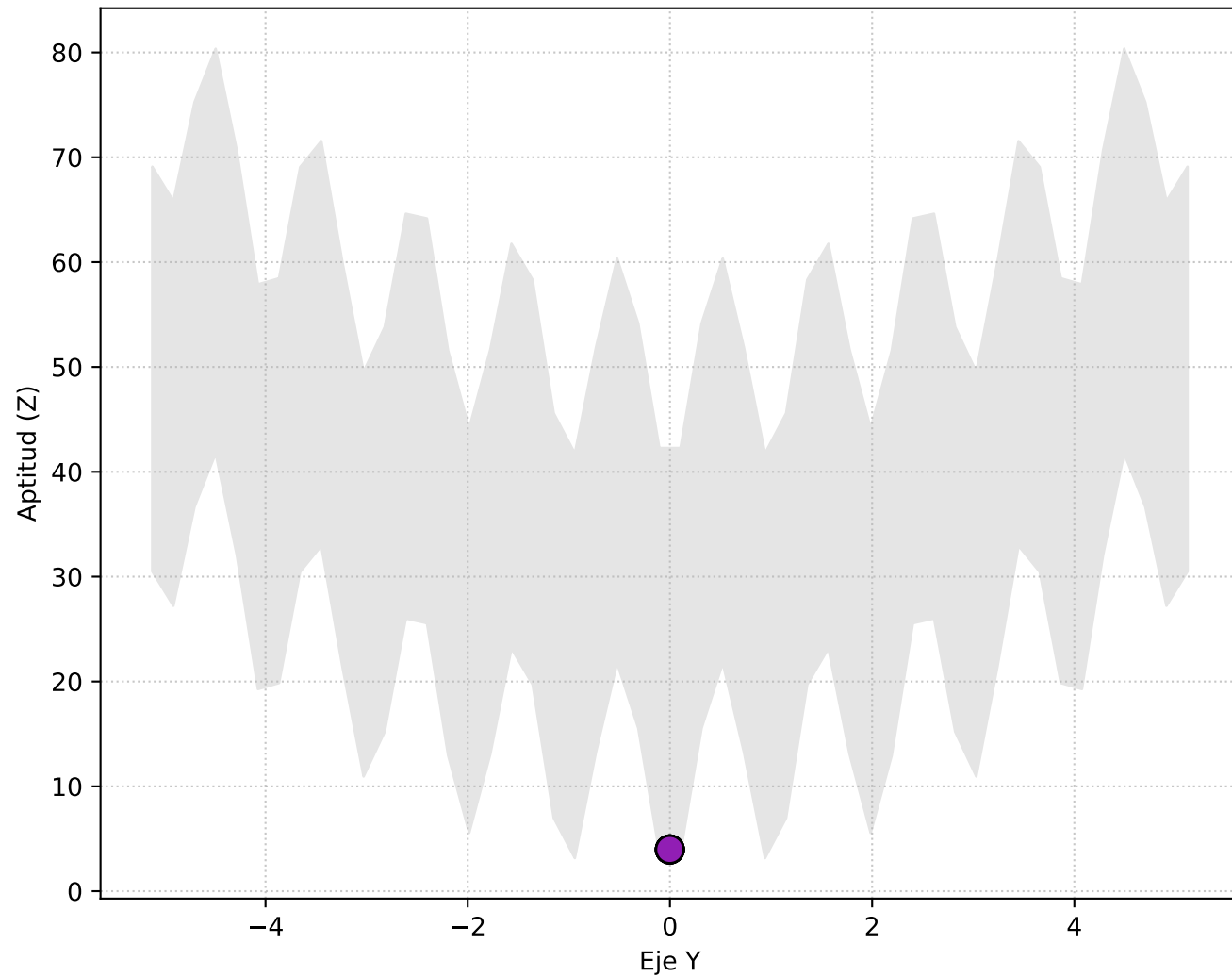
Perspectiva 3D



Vista Lateral (X vs Elevación Z)

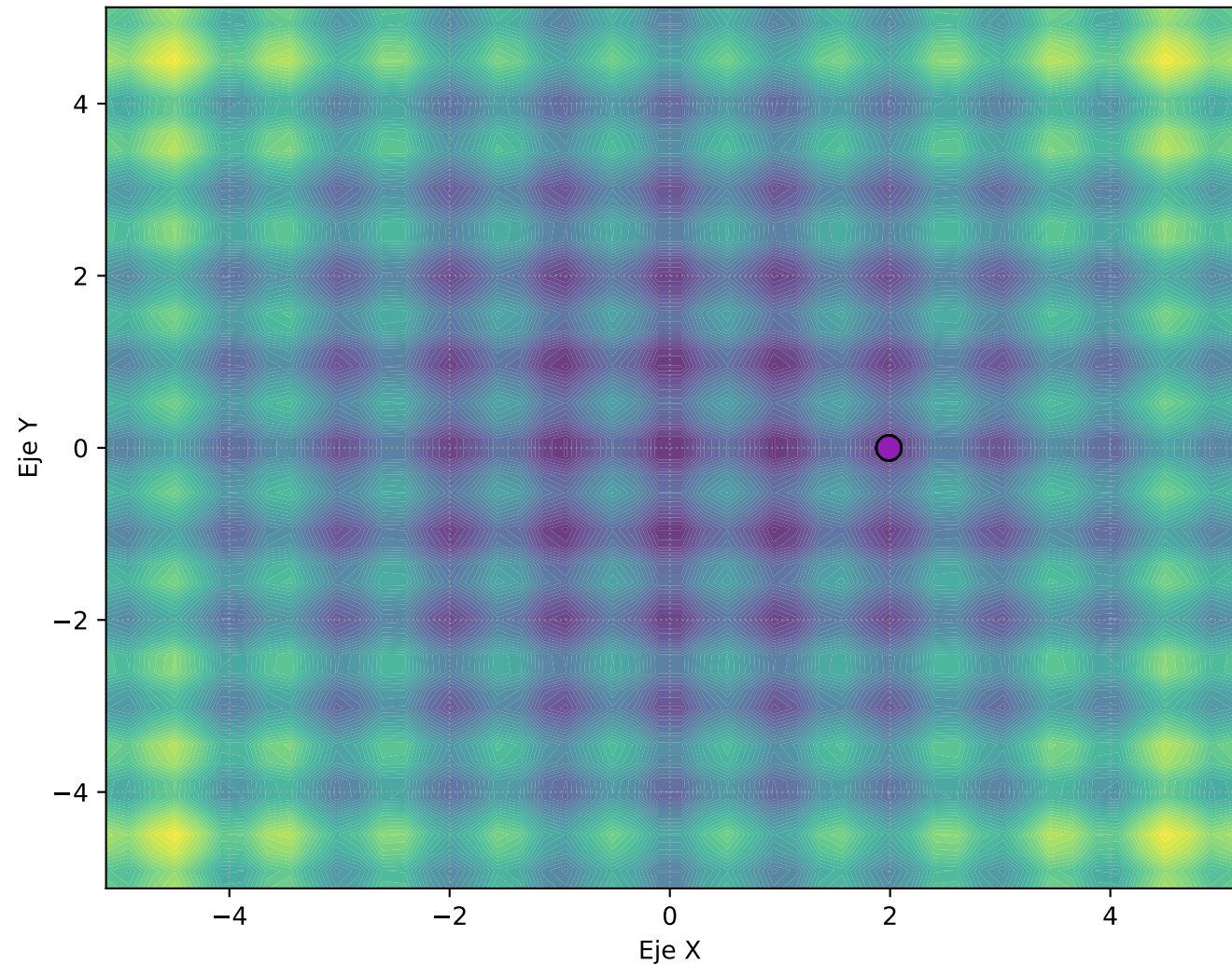


Vista Frontal (Y vs Elevación Z)

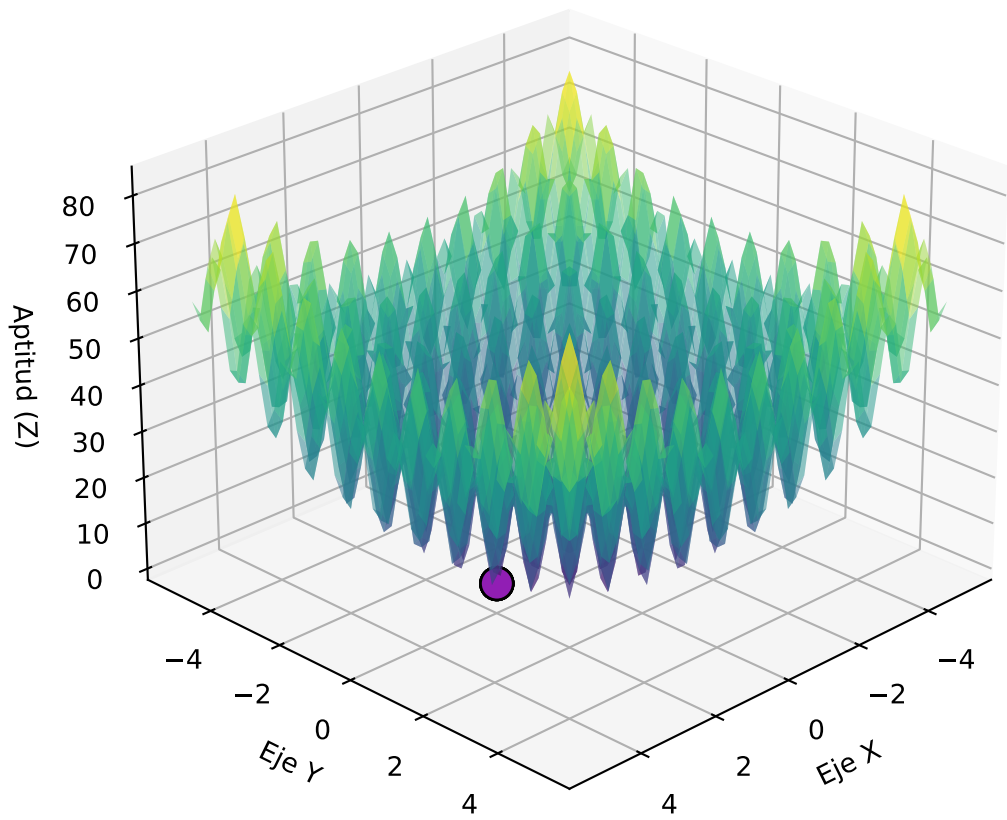


Progreso: Generación 70

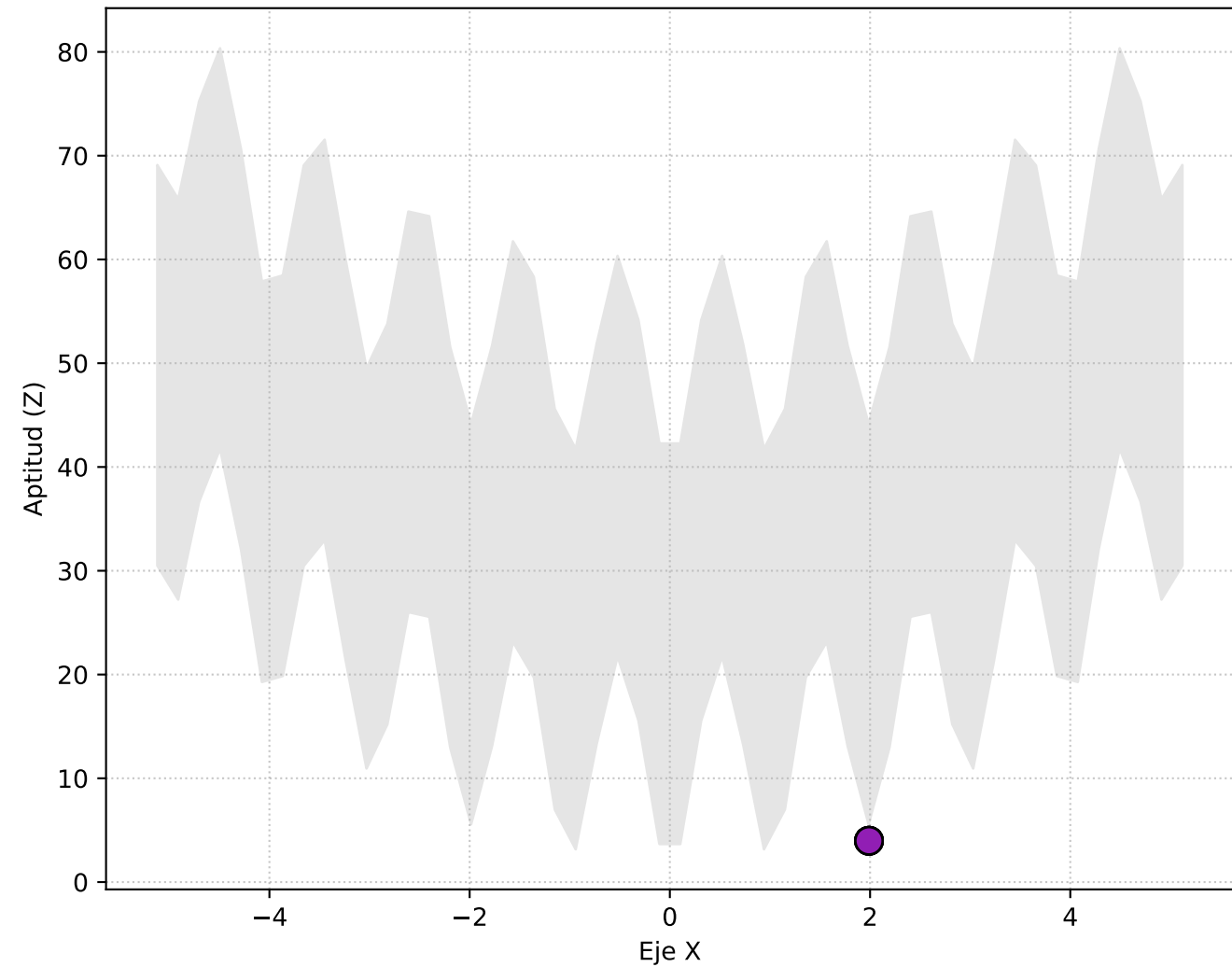
Vista Superior (X, Y)



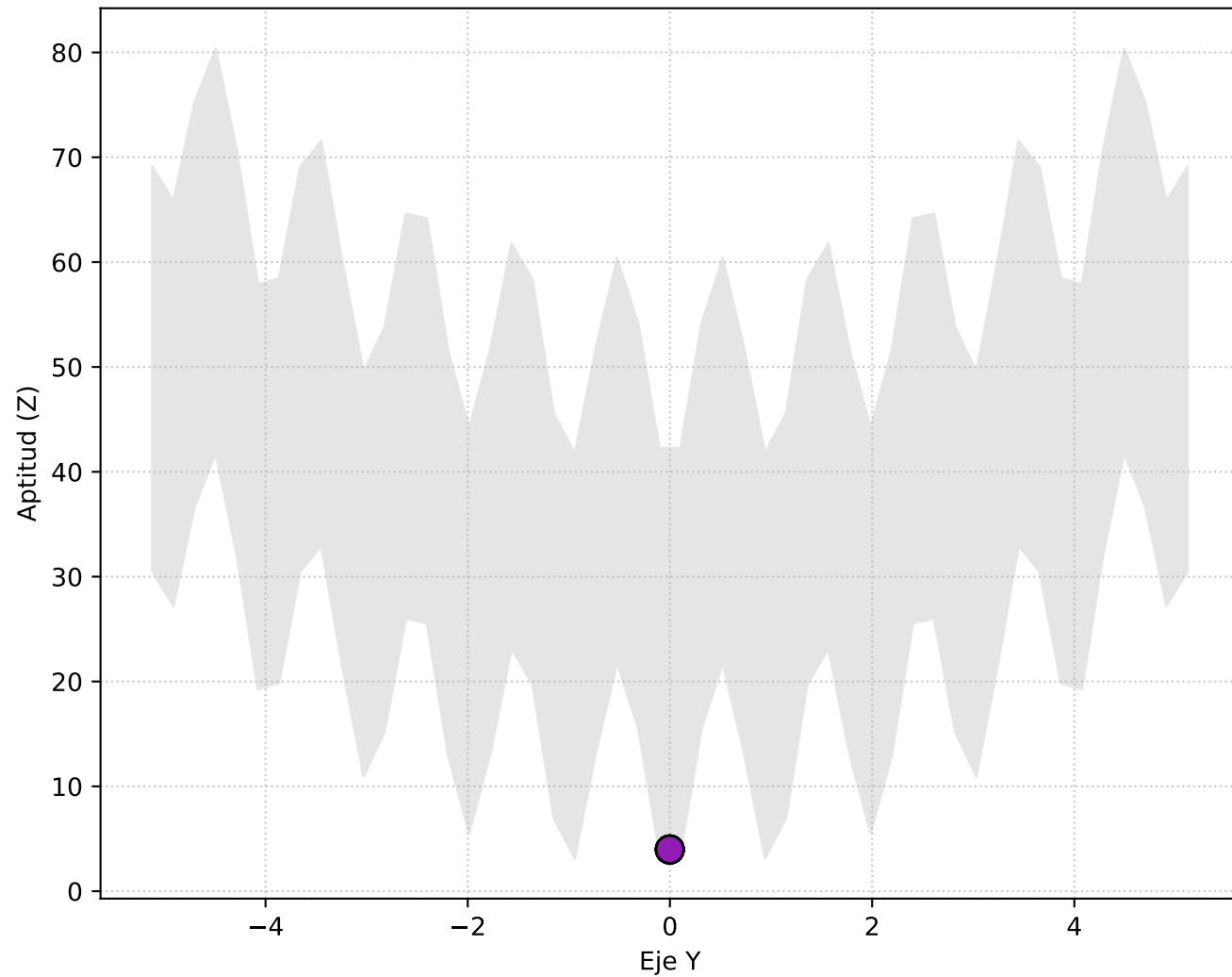
Perspectiva 3D



Vista Lateral (X vs Elevación Z)

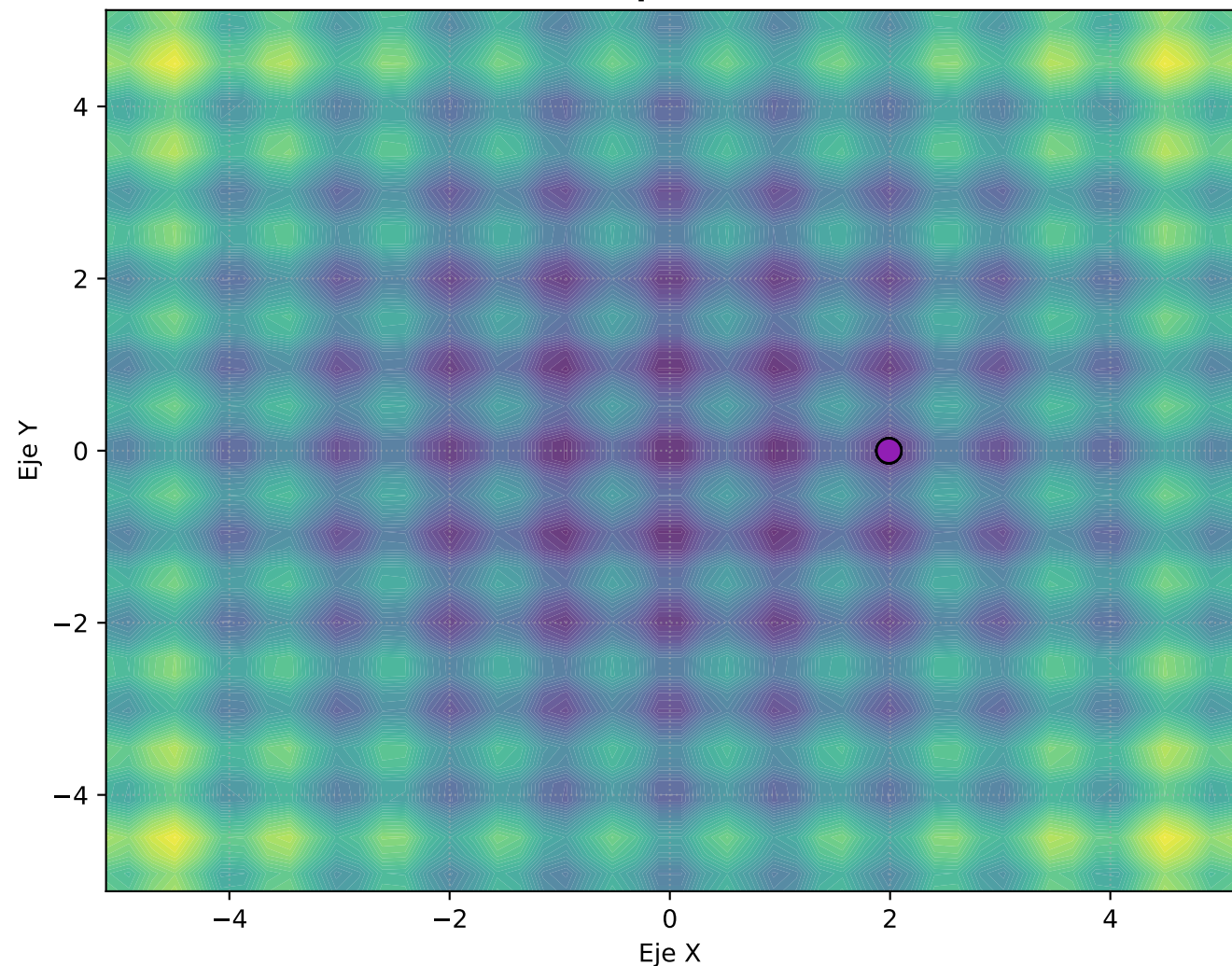


Vista Frontal (Y vs Elevación Z)

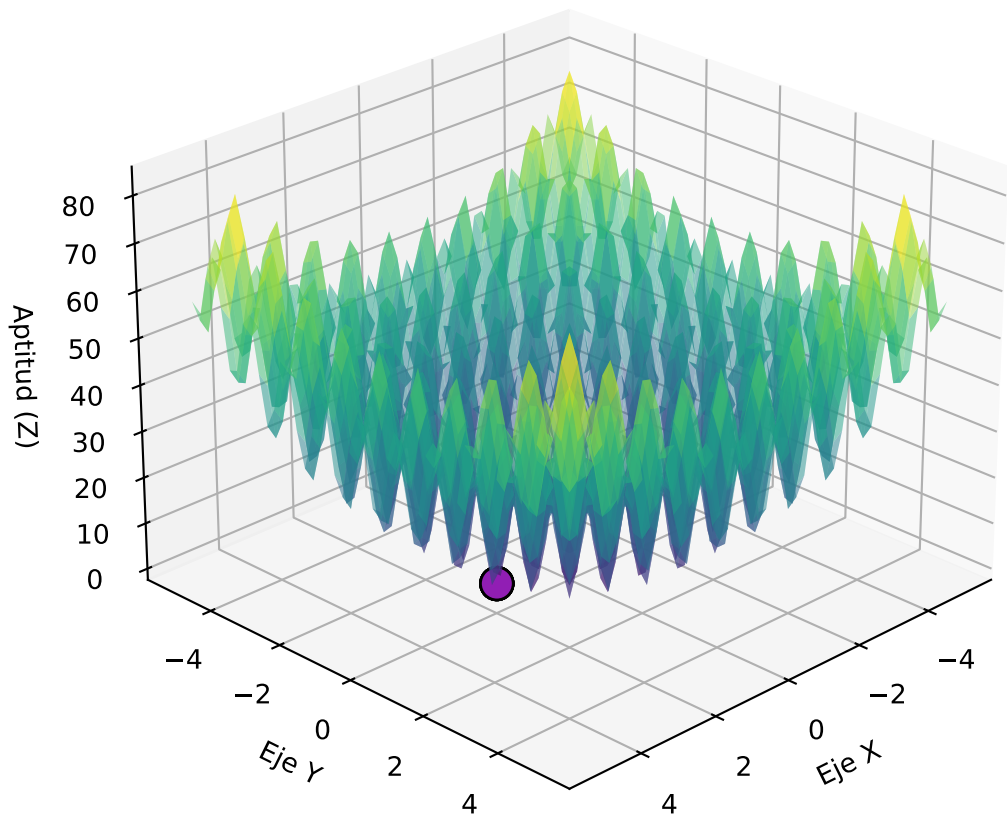


Progreso: Generación 80

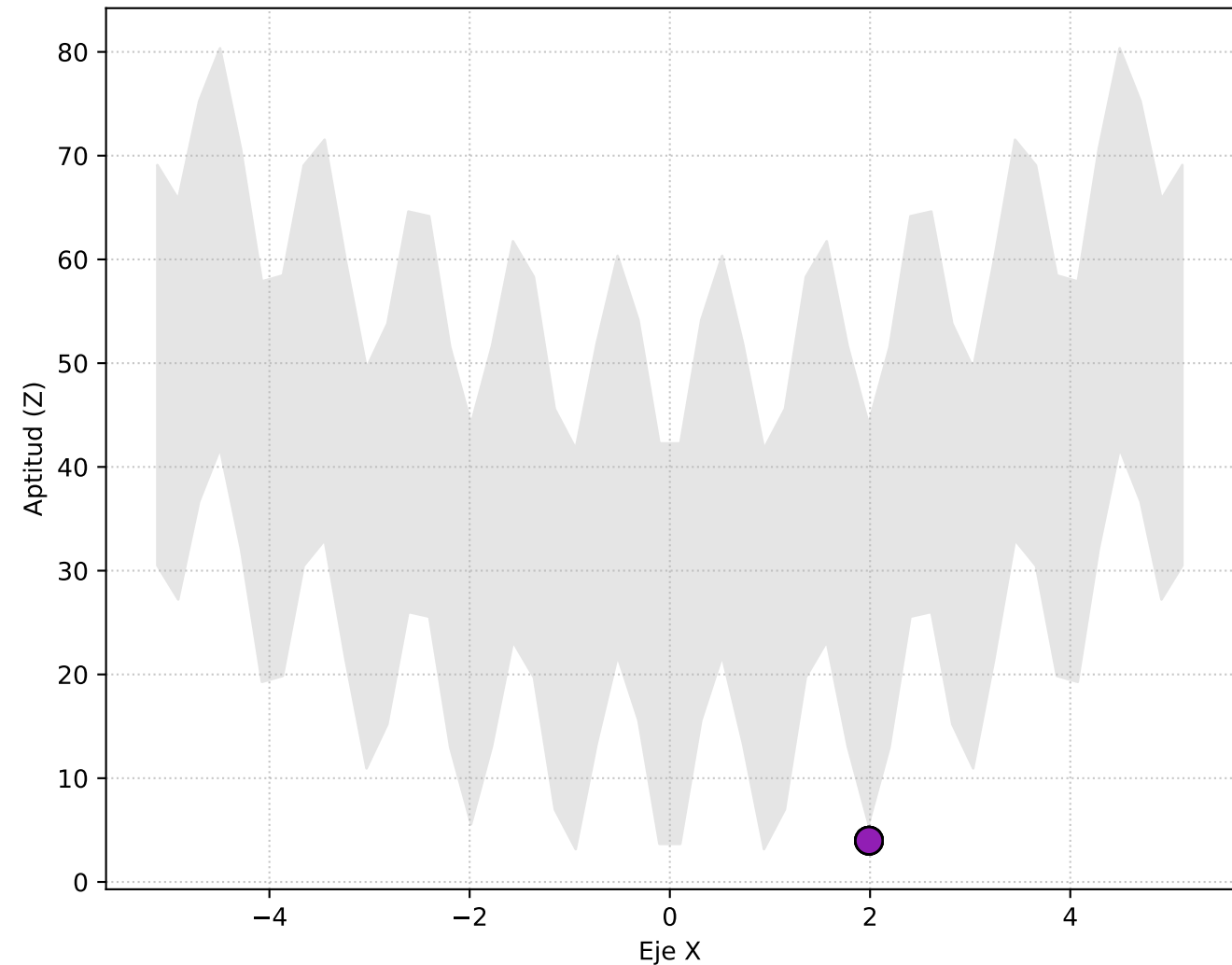
Vista Superior (X, Y)



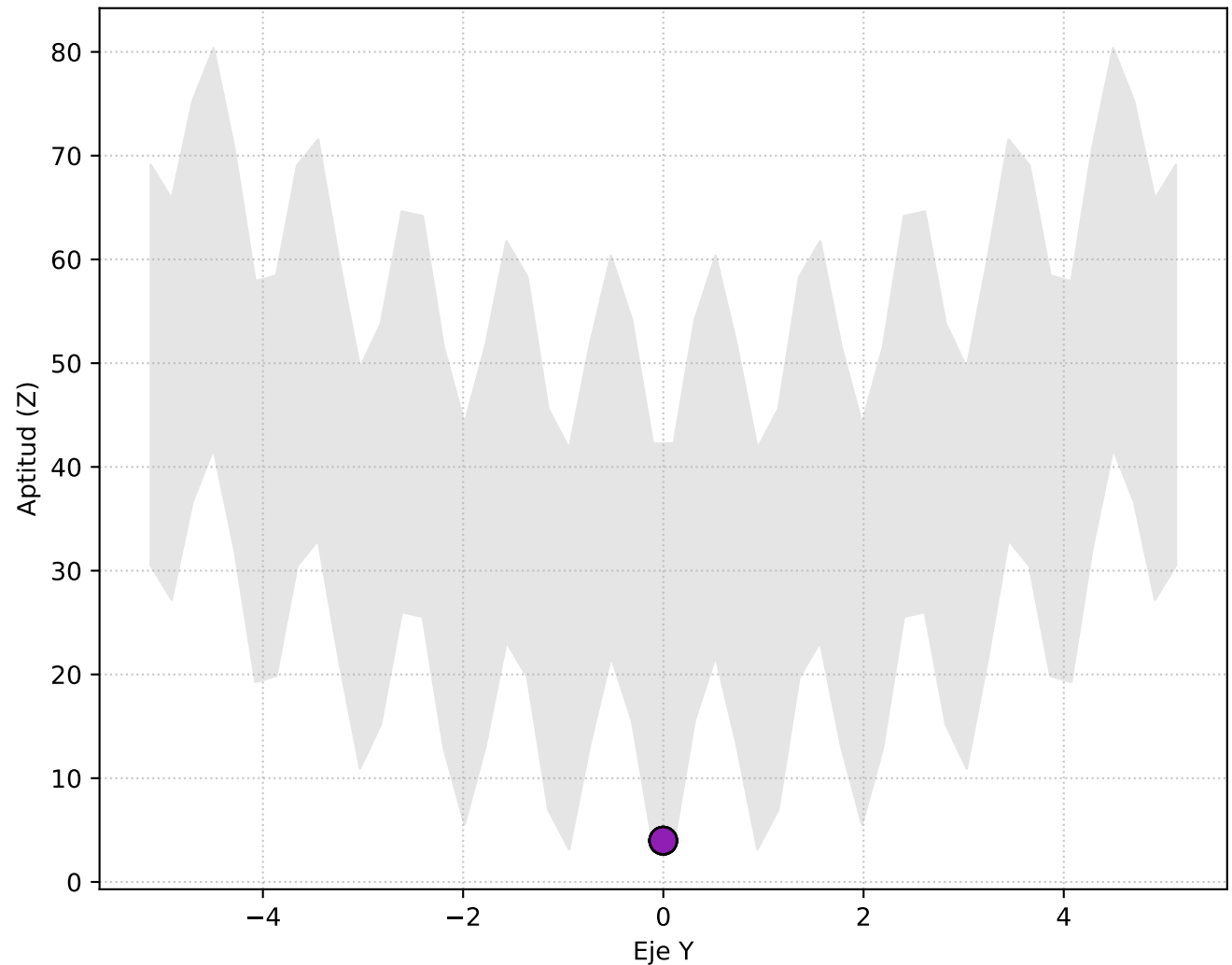
Perspectiva 3D



Vista Lateral (X vs Elevación Z)

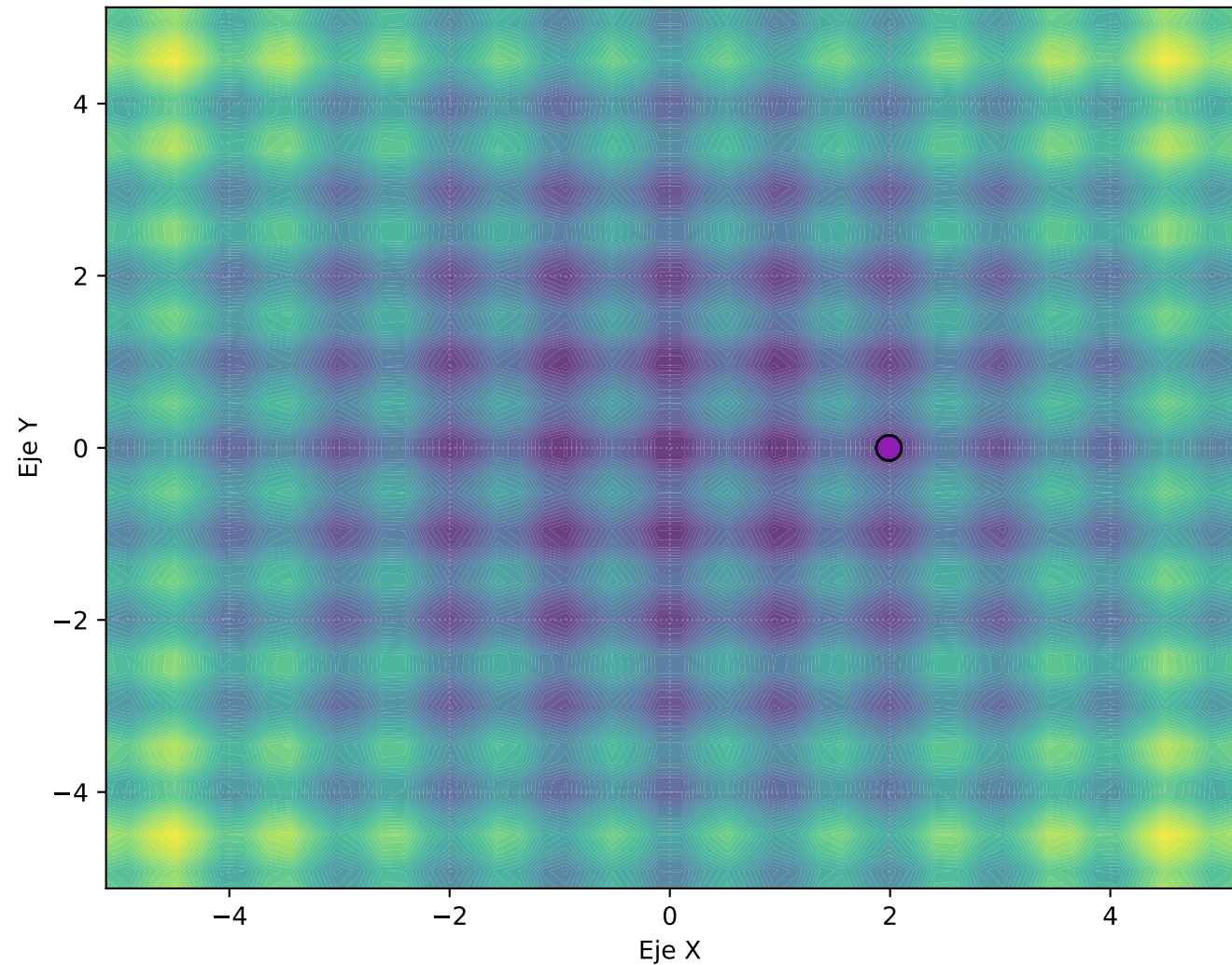


Vista Frontal (Y vs Elevación Z)

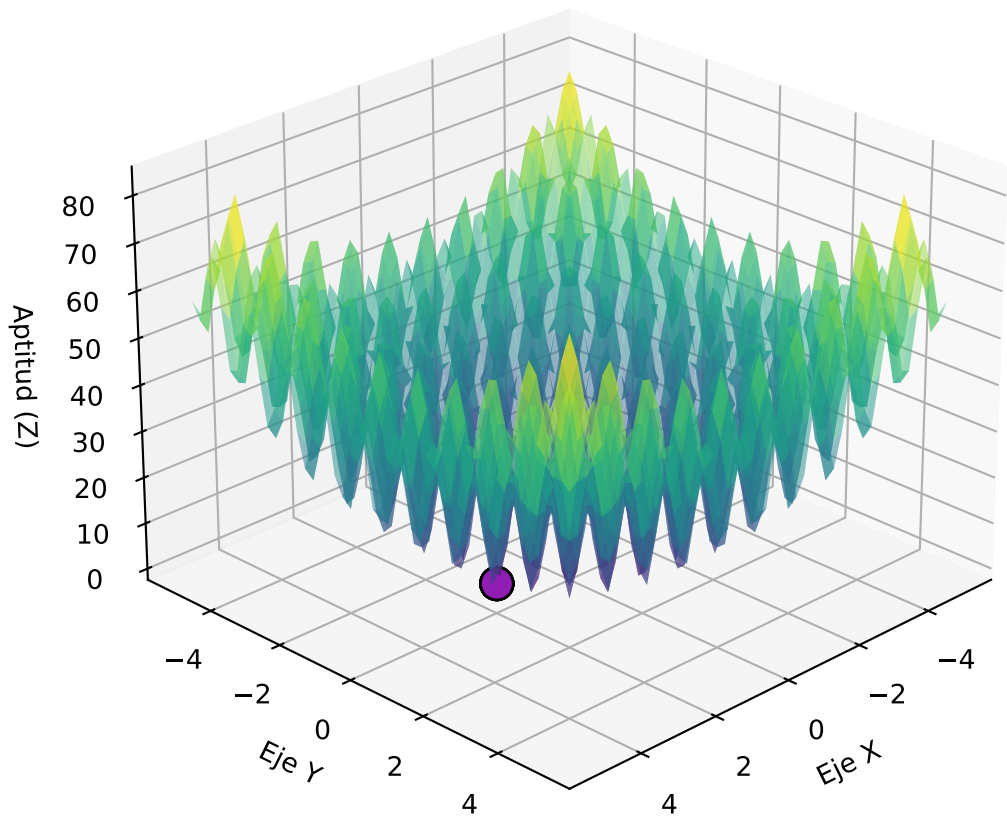


Progreso: Generación 90

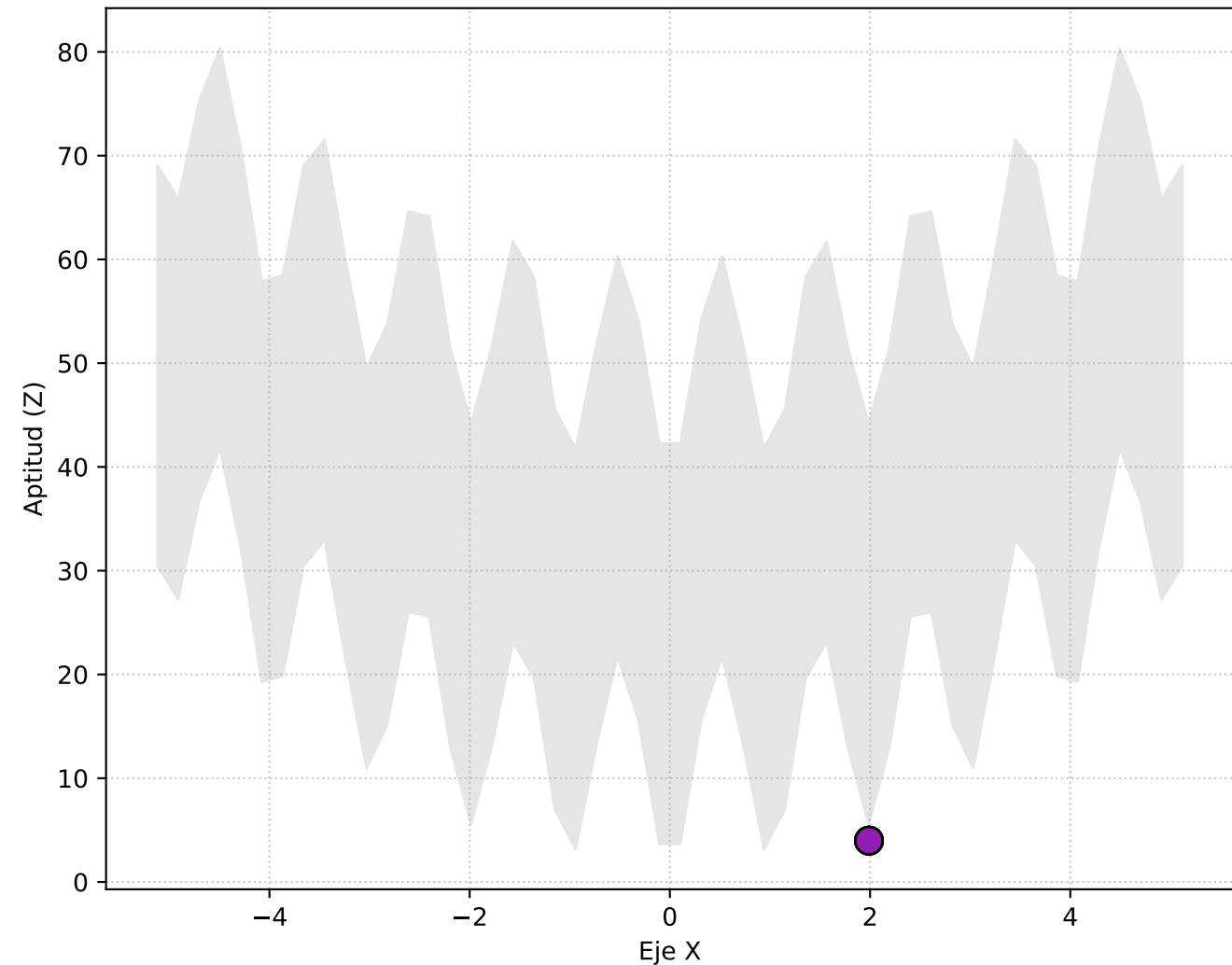
Vista Superior (X, Y)



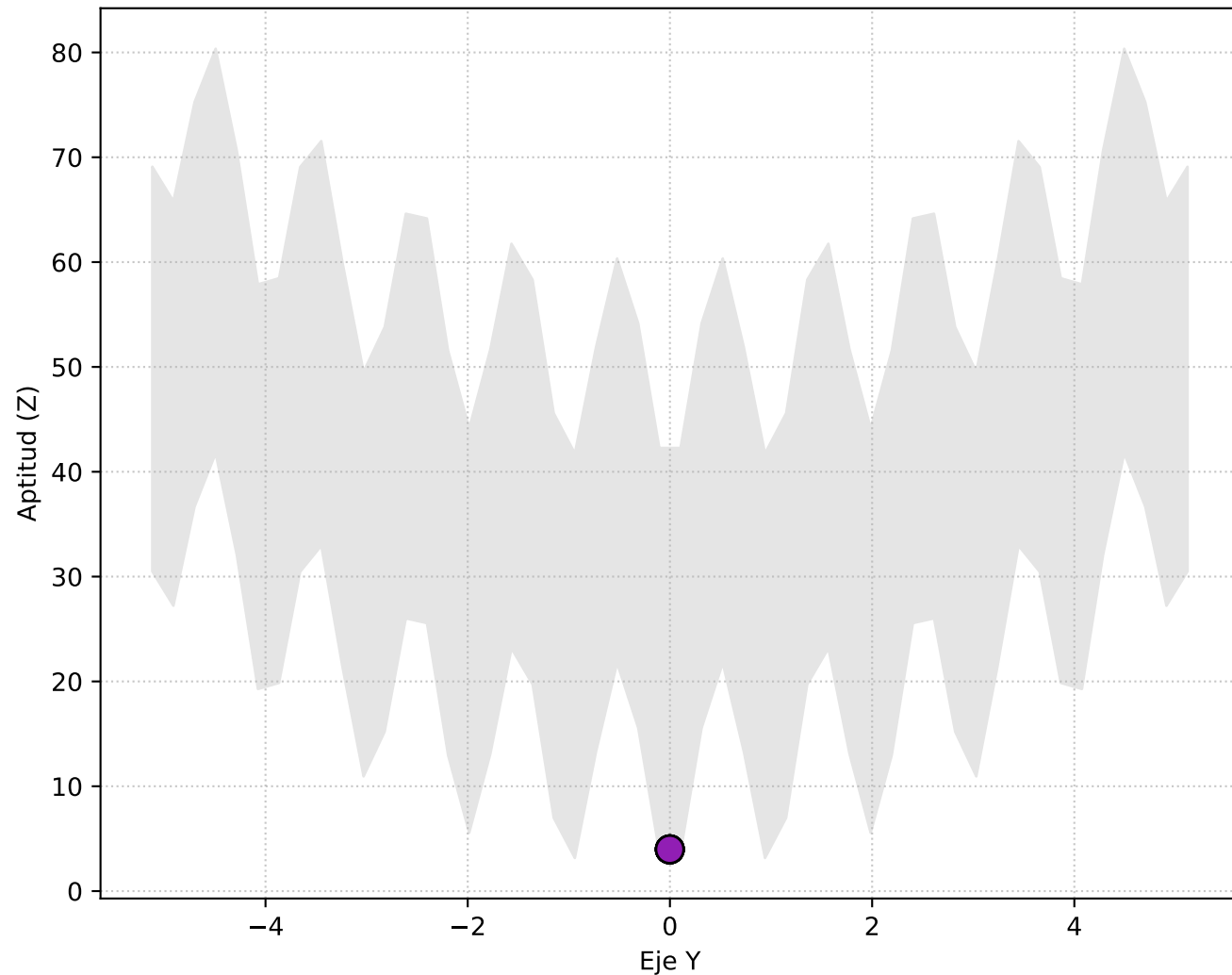
Perspectiva 3D



Vista Lateral (X vs Elevación Z)

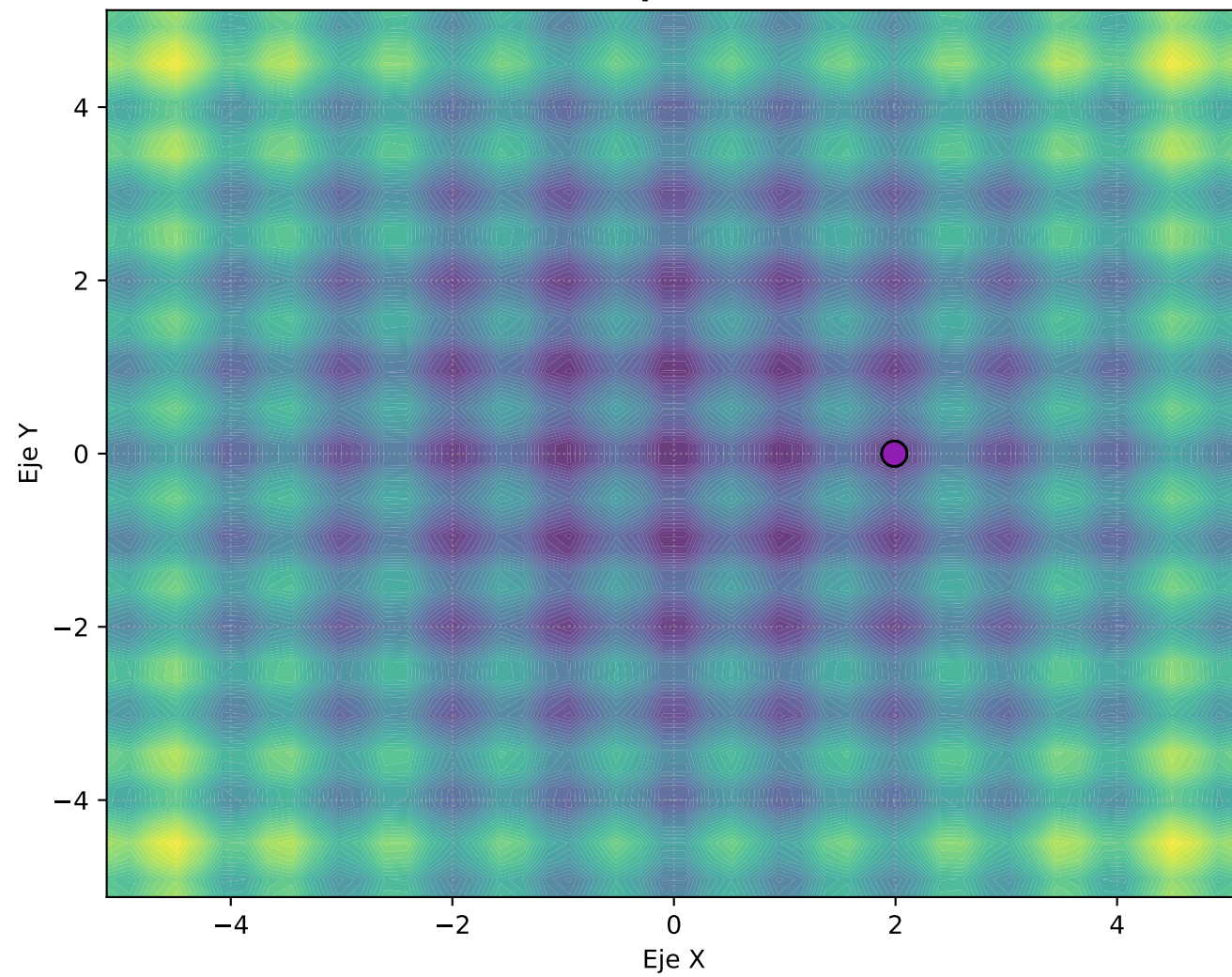


Vista Frontal (Y vs Elevación Z)

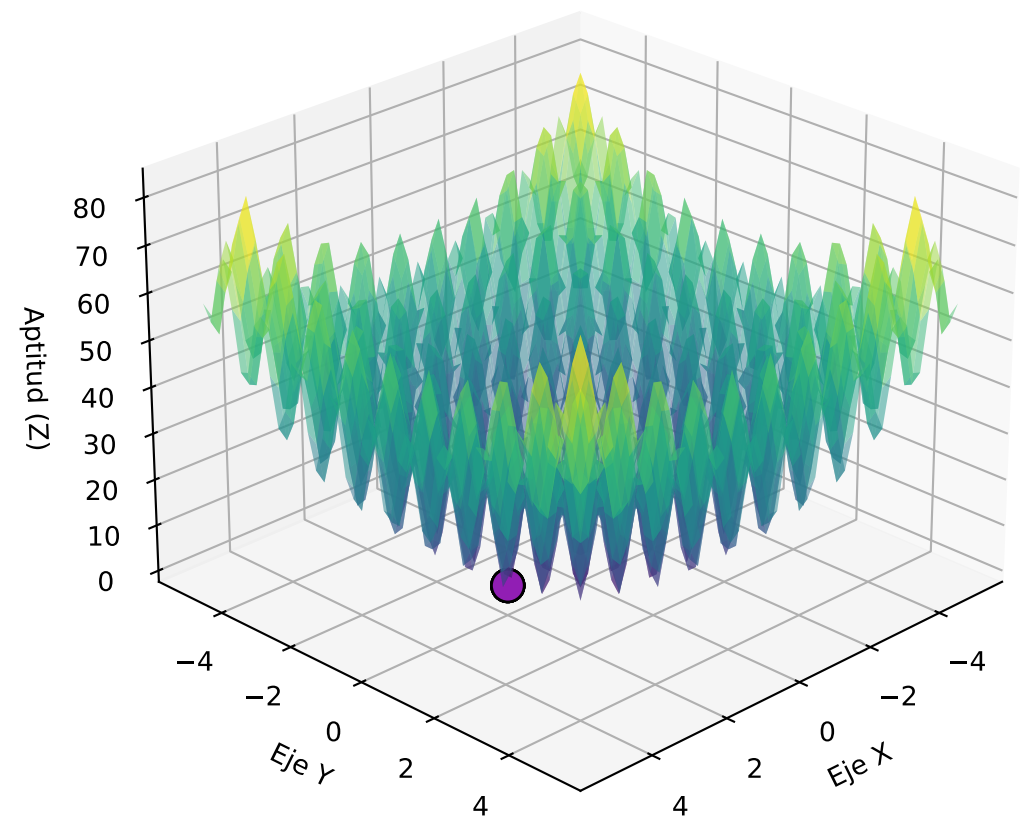


Progreso: Generación 100

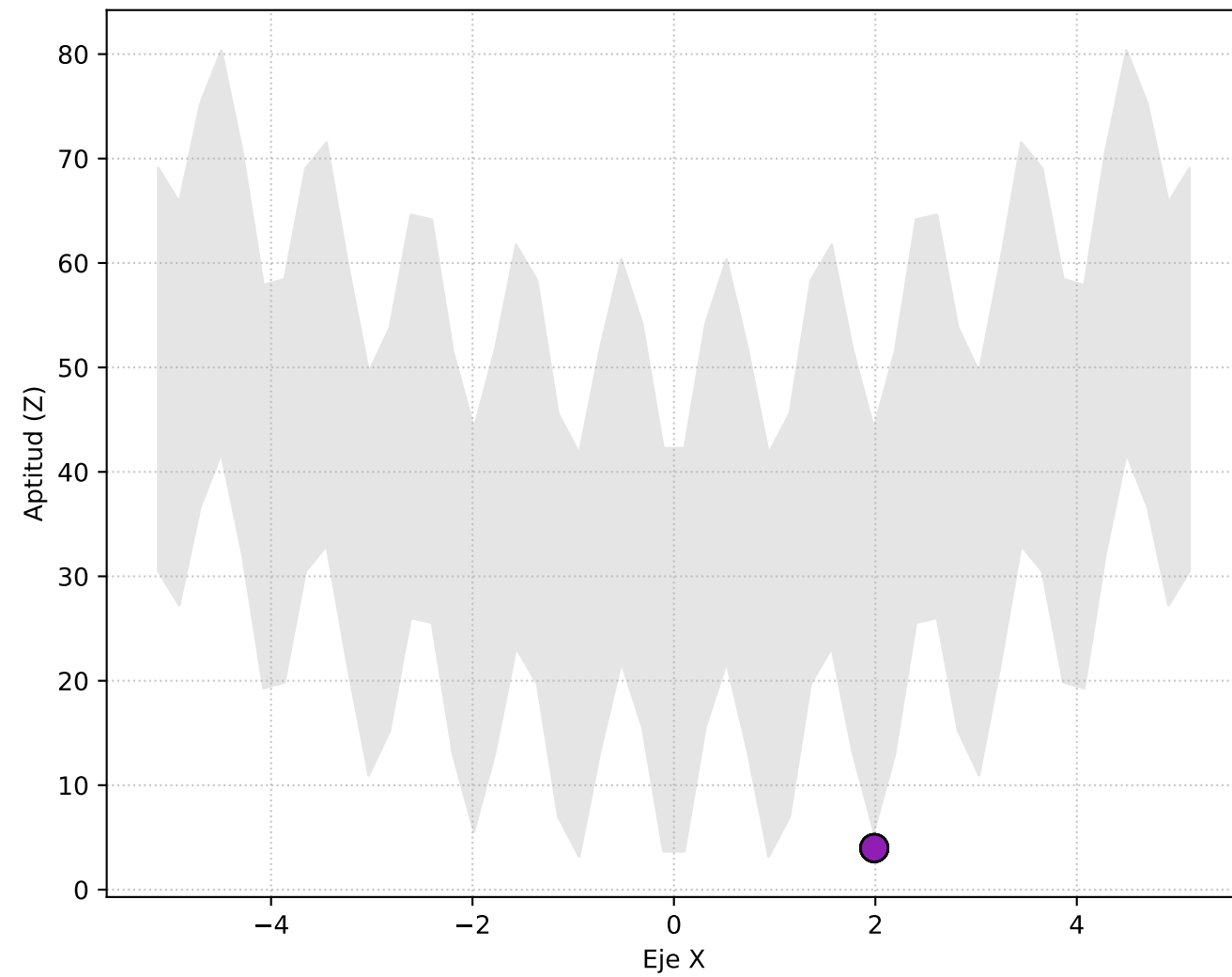
Vista Superior (X, Y)



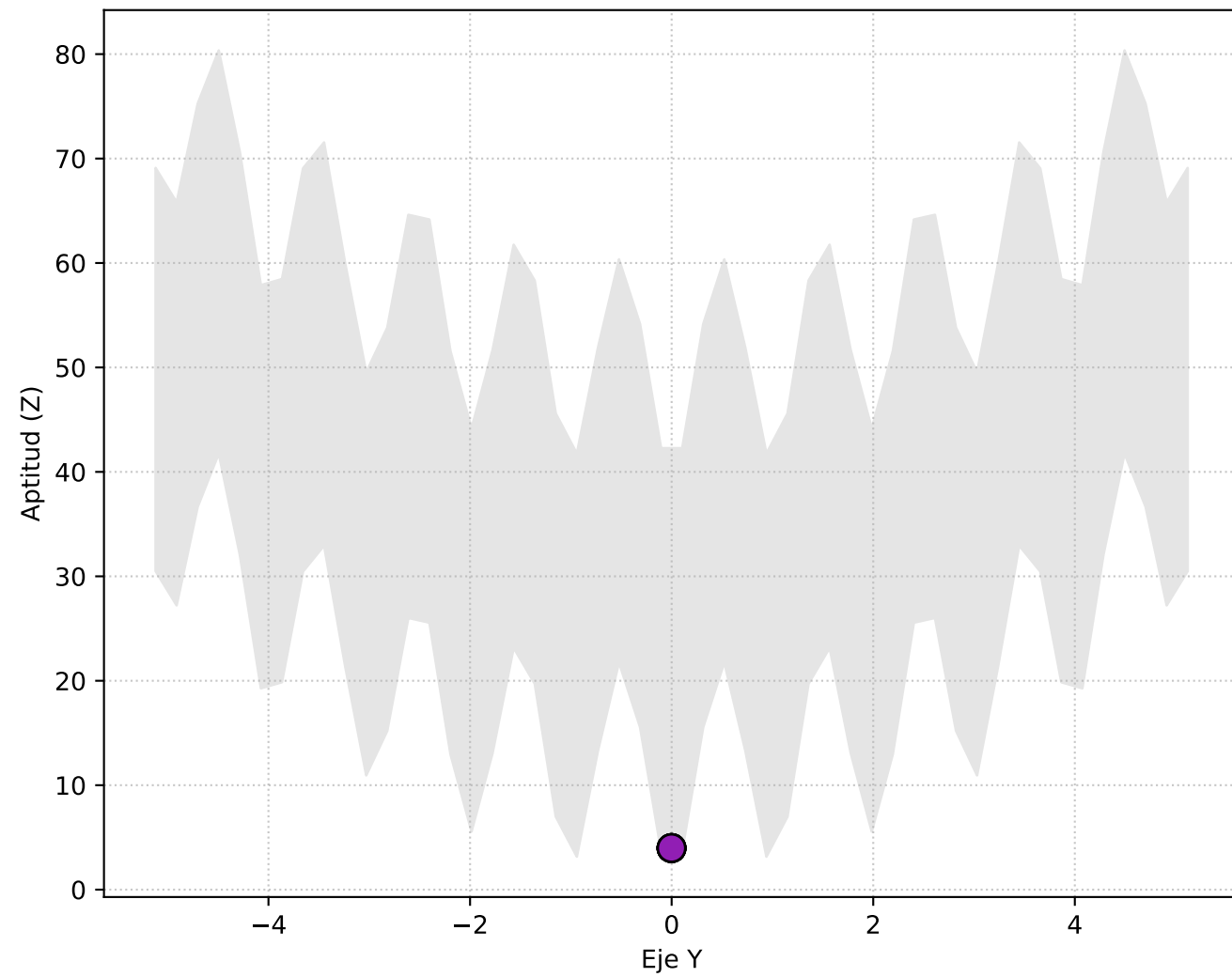
Perspectiva 3D



Vista Lateral (X vs Elevación Z)



Vista Frontal (Y vs Elevación Z)

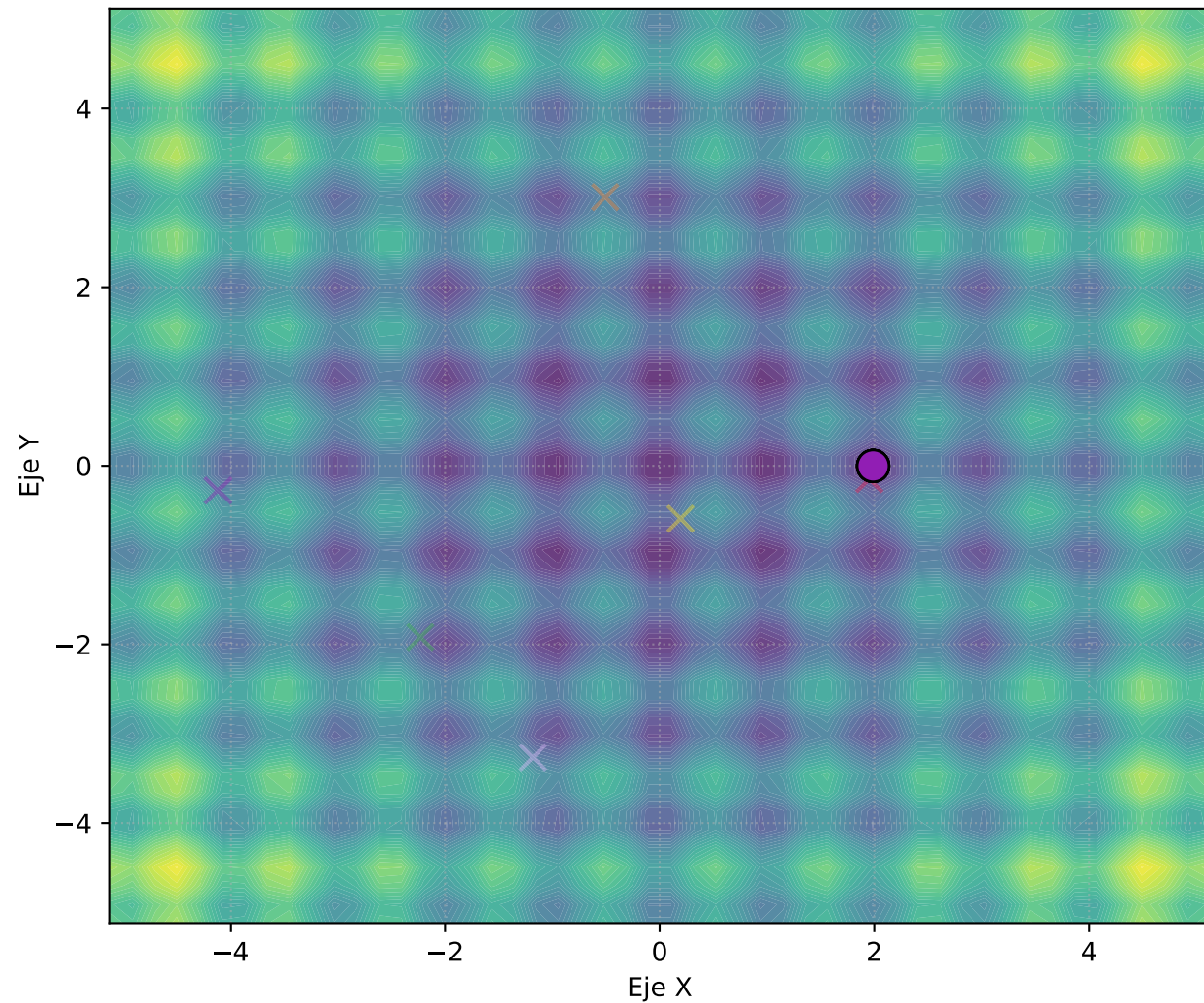


Posición del Mejor Individuo por Salto

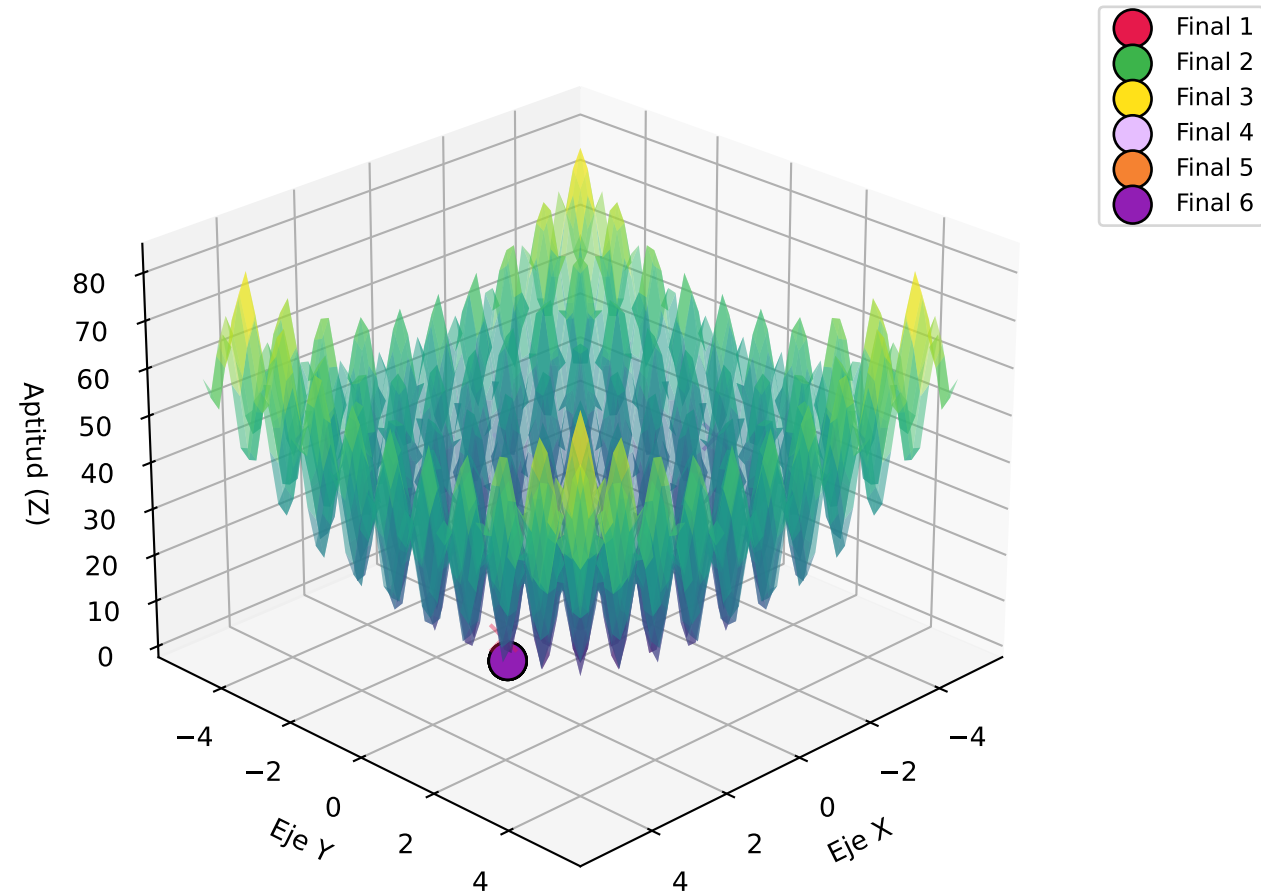
Gen	Mejores Bits	Coordenadas (X, Y)	Aptitud
10	[1, 1, 1, 0, 1, 0, 0, 0, 1, 0, 1, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 1, 1, 0]	X: 1.95, Y: -0.01	0.1903
20	[1, 1, 1, 0, 1, 0, 0, 0, 1, 0, 1, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0]	X: 1.95, Y: -0.00	0.1912
30	[1, 1, 1, 0, 1, 0, 0, 0, 1, 1, 1, 1, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0]	X: 1.97, Y: -0.00	0.1989
40	[1, 1, 1, 0, 1, 0, 0, 0, 1, 0, 1, 1, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0]	X: 1.99, Y: -0.00	0.2007
50	[1, 1, 1, 0, 1, 0, 0, 0, 1, 0, 1, 1, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0]	X: 1.99, Y: -0.00	0.2007
60	[1, 1, 1, 0, 1, 0, 0, 0, 1, 0, 1, 1, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0]	X: 1.99, Y: -0.00	0.2008
70	[1, 1, 1, 0, 1, 0, 0, 0, 1, 0, 1, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0]	X: 1.99, Y: -0.00	0.2008
80	[1, 1, 1, 0, 1, 0, 0, 0, 1, 0, 1, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0]	X: 1.99, Y: -0.00	0.2008
90	[1, 1, 1, 0, 1, 0, 0, 0, 1, 0, 1, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0]	X: 1.99, Y: -0.00	0.2008
100	[1, 1, 1, 0, 1, 0, 0, 0, 1, 0, 1, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0]	X: 1.99, Y: -0.00	0.2008

Comparativa: Iniciales (Tache) vs Finales Gen 100 (Círculo)

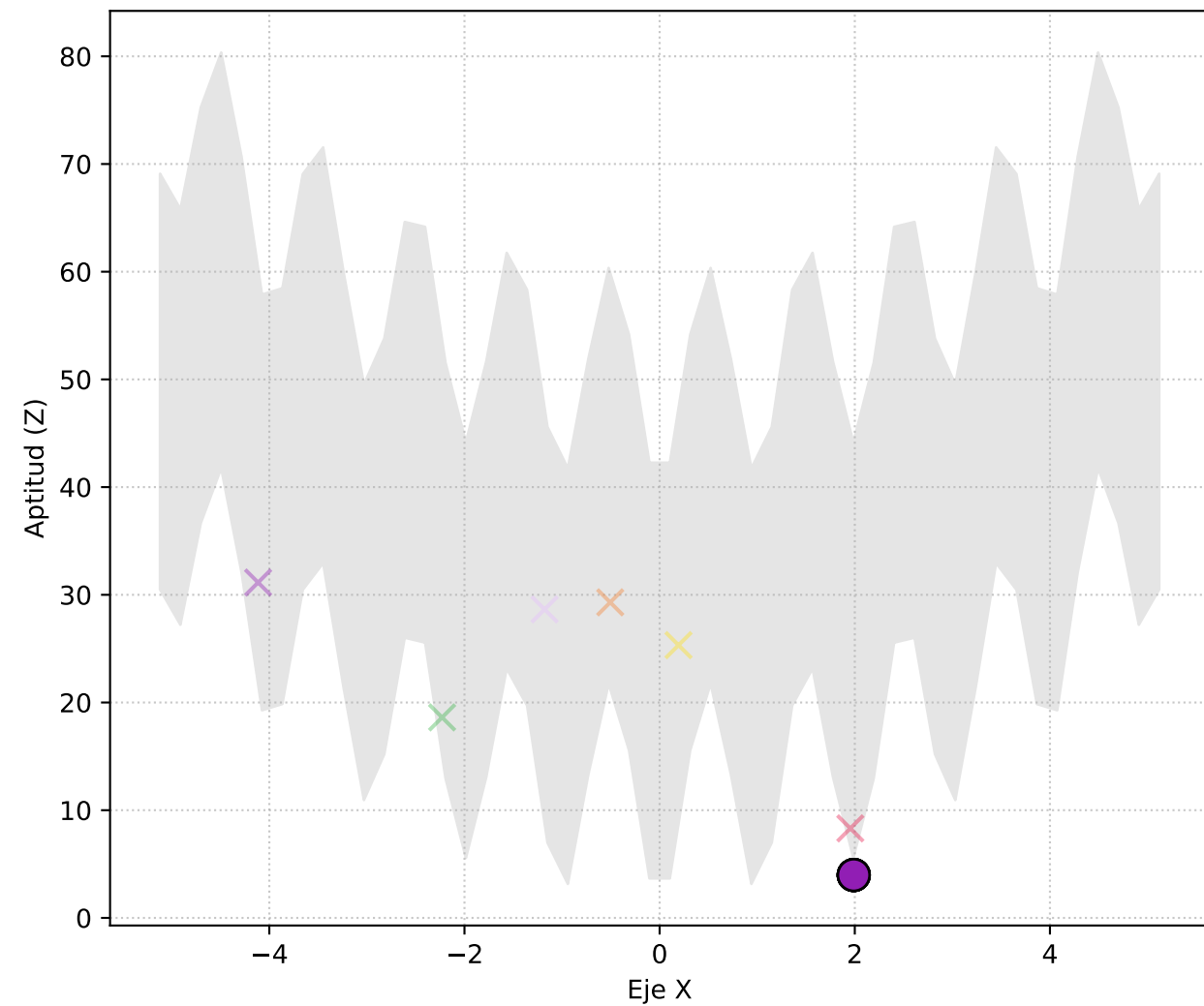
Vista Superior (X, Y)



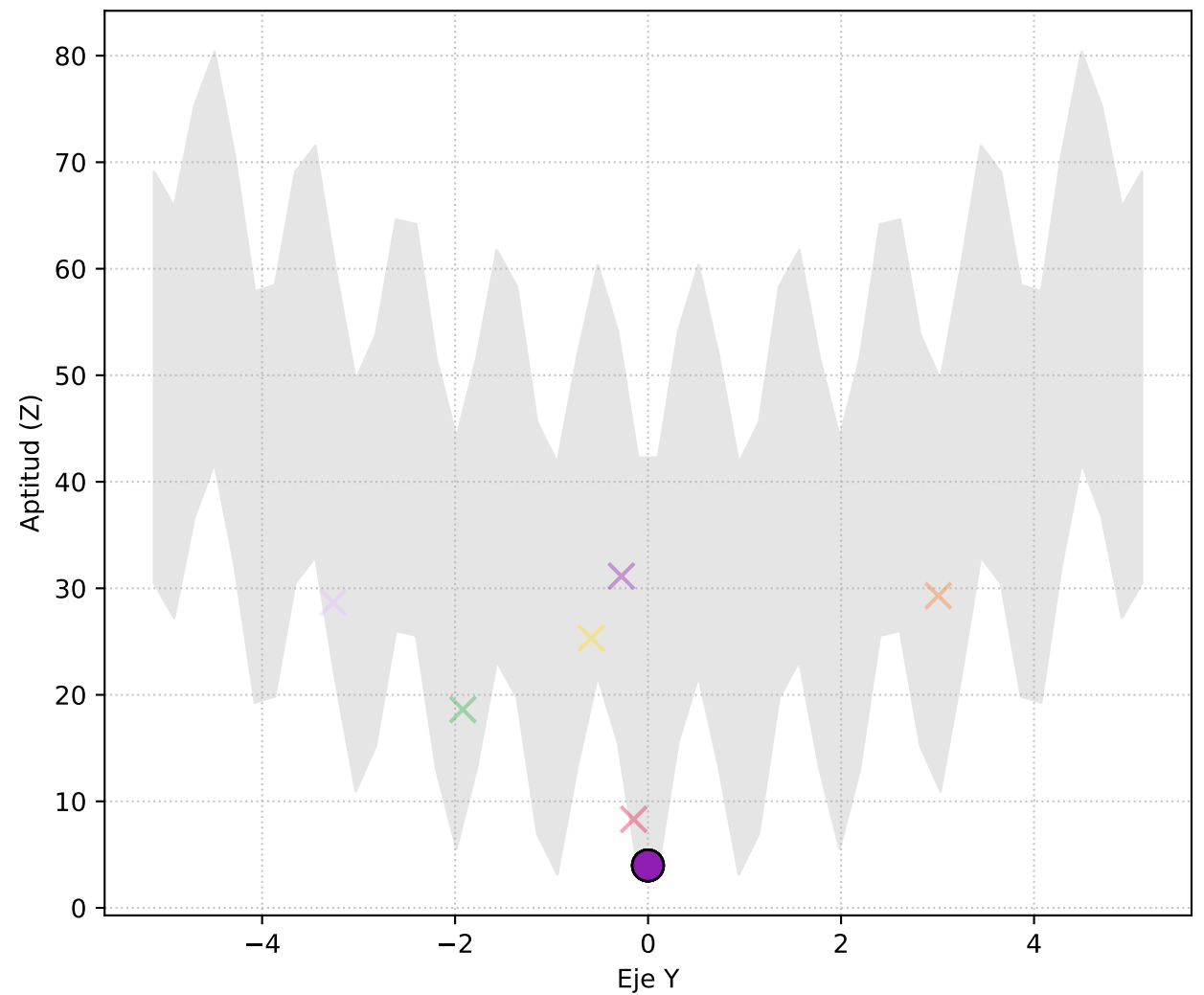
Perspectiva 3D



Vista Lateral (X vs Elevación Z)



Vista Frontal (Y vs Elevación Z)



--- GENERACIÓN 1 ---

1. Evaluación:

Bits: [0, 1, 0, 1, 0, 0, 1, 1, 1, 1, 0, 0, 0, 0, 1, 1, 1, 0, 0, 1, 0, 1, 0, 1] | Dec: (-1.18, -3.26) | Aptitud: 0.0
Bits: [1, 0, 0, 1, 0, 1, 0, 1, 1, 0, 0, 0, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0, 1, 0] | Dec: (4.12, -1.27) | Aptitud: 0.0
Bits: [1, 1, 0, 0, 0, 1, 1, 0, 1, 0, 1, 1, 0, 1, 0, 0, 1, 0, 0, 1, 1, 0, 1, 0] | Dec: (0.19, -0.59) | Aptitud: 0.0
Bits: [1, 0, 0, 0, 0, 0, 1, 1, 1, 1, 0, 1, 0, 0, 0, 0, 1, 1, 1, 1, 0, 0, 1, 1] | Dec: (5.02, -4.71) | Aptitud: 0.0
Bits: [0, 1, 0, 0, 1, 0, 1, 0, 1, 1, 1, 1, 1, 0, 1, 0, 1, 1, 1, 0, 1, 0, 1, 1] | Dec: (-0.51, 3.01) | Aptitud: 0.0
Bits: [0, 1, 1, 0, 1, 1, 0, 0, 0, 1, 1, 0, 0, 1, 1, 1, 1, 0, 0, 0, 0, 0, 0, 0] | Dec: (-2.23, -1.92) | Aptitud: 0.0
Bits: [1, 1, 1, 0, 1, 0, 0, 0, 1, 0, 1, 1, 0, 1, 0, 0, 0, 0, 1, 0, 0, 1, 1, 0] | Dec: (1.95, -0.15) | Aptitud: 0.1
Bits: [0, 0, 0, 1, 0, 1, 0, 1, 1, 0, 1, 1, 0, 1, 0, 0, 0, 1, 0, 1, 1, 0, 0, 1] | Dec: (-4.11, -0.28) | Aptitud: 0.0
Bits: [0, 0, 1, 0, 0, 0, 1, 1, 1, 1, 1, 1, 0, 0, 1, 0, 0, 1, 1, 1, 0, 0, 0, 0] | Dec: (-2.67, -2.80) | Aptitud: 0.0
Bits: [1, 0, 1, 1, 1, 1, 0, 1, 0, 0, 1, 0, 0, 1, 1, 1, 0, 0, 1, 0, 1, 0, 1, 0] | Dec: (3.45, -1.41) | Aptitud: 0.0

2. Selección:

[1, 1, 1, 0, 1, 0, 0, 0, 1, 0, 1, 1, 0, 1, 0, 0, 0, 0, 1, 0, 0, 1, 1, 0]
[1, 1, 1, 0, 1, 0, 0, 0, 1, 0, 1, 1, 0, 1, 0, 0, 0, 0, 1, 0, 0, 1, 1, 0]
[1, 1, 0, 0, 0, 1, 1, 0, 1, 0, 1, 1, 0, 1, 0, 0, 1, 0, 0, 1, 1, 0, 1, 0]
[1, 1, 0, 0, 0, 1, 1, 0, 1, 0, 1, 1, 0, 1, 0, 0, 1, 0, 0, 1, 1, 0, 1, 0]
[0, 1, 0, 0, 1, 0, 1, 0, 1, 1, 1, 1, 0, 1, 0, 1, 1, 0, 1, 0, 1, 0, 1, 1]

3. Cruza:

Cruzando [1, 1, 1, 0, 1, 0, 0, 0, 1, 0, 1, 1, 0, 1, 0, 0, 0, 0, 1, 0, 0, 1, 1, 0] y [1, 1, 1, 0, 1, 0, 0, 0, 1, 0, 0, 0, 1, 0, 0, 0, 1, 1, 0, 1, 0, 0, 1, 0]
Cruzando [1, 1, 1, 0, 1, 0, 0, 0, 1, 0, 1, 1, 0, 1, 0, 0, 0, 0, 1, 0, 0, 1, 1, 0] y [1, 1, 0, 0, 0, 1, 1, 0, 1, 0, 0, 0, 0, 1, 1, 0, 1, 0, 0, 1, 0, 0, 1, 0]
Cruzando [1, 1, 0, 0, 0, 1, 1, 0, 1, 0, 1, 1, 0, 1, 0, 0, 1, 0, 0, 1, 1, 0, 1, 0] y [1, 1, 0, 0, 0, 1, 1, 0, 1, 0, 0, 0, 1, 1, 0, 1, 0, 1, 0, 1, 0, 0, 1, 0]
Cruzando [1, 1, 0, 0, 0, 1, 1, 0, 1, 0, 1, 1, 0, 1, 0, 0, 1, 0, 0, 1, 1, 0, 1, 0] y [0, 1, 0, 0, 1, 0, 1, 0, 1, 0, 1, 1, 0, 1, 0, 1, 0, 1, 0, 1, 0, 0, 1, 0]
Cruzando [0, 1, 0, 0, 1, 0, 1, 0, 1, 1, 1, 1, 1, 0, 1, 0, 1, 1, 1, 0, 1, 0, 1, 1] y [1, 1, 1, 0, 1, 0, 0, 0, 1, 0, 0, 0, 1, 0, 0, 1, 0, 1, 0, 1, 0, 0, 1, 0]

4. Mutación:

¡Mutación! [1, 1, 1, 0, 1, 0, 0, 0, 1, 0, 1, 1, 0, 1, 0, 0, 1, 0, 0, 1, 1, 0, 1, 0] -> [1, 1, 1, 0, 1, 0, 0, 0, 1, 0, 0, 0, 1, 0, 0, 1, 1, 0, 1, 0, 1, 0, 0, 1]
¡Mutación! [1, 1, 0, 0, 0, 1, 1, 0, 1, 0, 1, 1, 0, 1, 0, 0, 1, 0, 0, 1, 1, 0, 1, 0] -> [1, 1, 0, 0, 0, 1, 1, 0, 1, 0, 1, 0, 0, 1, 0, 0, 1, 1, 0, 1, 0, 1, 0, 1]

--- GENERACIÓN 2 ---

1. Evaluación:

Bits: [1, 1, 1, 0, 1, 0, 0, 0, 1, 0, 1, 1, 0, 1, 0, 0, 0, 0, 1, 0, 0, 1, 1, 0] | Dec: (1.95, -0.15) | Aptitud: 0.1
Bits: [1, 1, 1, 0, 1, 0, 0, 0, 1, 0, 1, 1, 0, 1, 0, 0, 0, 0, 1, 0, 0, 1, 1, 0] | Dec: (1.95, -0.15) | Aptitud: 0.1
Bits: [1, 1, 1, 0, 1, 0, 0, 0, 1, 0, 1, 0, 0, 1, 0, 0, 1, 0, 0, 1, 1, 0, 1, 0] | Dec: (1.95, -0.59) | Aptitud: 0.0
Bits: [1, 1, 0, 0, 0, 1, 1, 0, 1, 0, 1, 1, 0, 1, 0, 0, 0, 0, 1, 0, 0, 1, 1, 0] | Dec: (0.19, -0.15) | Aptitud: 0.0
Bits: [1, 1, 0, 0, 0, 1, 1, 0, 1, 0, 1, 1, 0, 0, 0, 0, 1, 0, 0, 1, 1, 0, 1, 0] | Dec: (0.19, -4.53) | Aptitud: 0.0
Bits: [1, 1, 0, 0, 0, 1, 1, 0, 1, 0, 1, 1, 0, 1, 0, 0, 1, 0, 0, 1, 1, 0, 1, 0] | Dec: (0.19, -0.59) | Aptitud: 0.0
Bits: [1, 1, 0, 0, 0, 1, 1, 0, 1, 0, 1, 1, 0, 0, 1, 0, 1, 1, 1, 0, 1, 0, 1, 1] | Dec: (0.19, -3.01) | Aptitud: 0.0
Bits: [0, 1, 0, 0, 1, 0, 1, 0, 1, 1, 1, 1, 1, 1, 0, 0, 1, 0, 0, 1, 1, 0, 1, 0] | Dec: (-0.51, 0.59) | Aptitud: 0.0
Bits: [0, 1, 0, 0, 1, 0, 1, 0, 1, 1, 1, 1, 1, 1, 0, 0, 0, 0, 1, 0, 0, 1, 1, 0] | Dec: (-0.51, 0.15) | Aptitud: 0.0
Bits: [1, 1, 1, 0, 1, 0, 0, 0, 1, 0, 1, 1, 0, 0, 1, 0, 1, 1, 1, 0, 1, 0, 1, 1] | Dec: (1.95, -3.01) | Aptitud: 0.0

2. Selección:

[1, 1, 1, 0, 1, 0, 0, 0, 1, 0, 1, 1, 0, 1, 0, 0, 0, 0, 1, 0, 0, 1, 1, 0]
[1, 1, 1, 0, 1, 0, 0, 0, 1, 0, 1, 1, 0, 1, 0, 0, 0, 0, 1, 0, 0, 1, 1, 0]
[1, 1, 1, 0, 1, 0, 0, 0, 1, 0, 1, 1, 0, 1, 0, 0, 0, 0, 1, 0, 0, 1, 1, 0]
[1, 1, 1, 0, 1, 0, 0, 0, 1, 0, 1, 1, 0, 1, 0, 0, 0, 0, 1, 0, 0, 1, 1, 0]
[1, 1, 1, 0, 1, 0, 0, 0, 1, 0, 1, 0, 0, 1, 0, 0, 1, 0, 0, 1, 1, 0, 1, 0]

3. Cruza:

Cruzando [1, 1, 1, 0, 1, 0, 0, 0, 1, 0, 1, 1, 0, 1, 0, 0, 0, 0, 1, 0, 0, 1, 1, 0] y [1, 1, 1, 0, 1, 0, 0, 0, 1, 0, 0, 0, 1, 0, 0, 0, 1, 0, 0, 1, 0, 0, 1, 0]
Cruzando [1, 1, 1, 0, 1, 0, 0, 0, 1, 0, 1, 1, 0, 1, 0, 0, 0, 0, 1, 0, 0, 1, 1, 0] y [1, 1, 1, 0, 1, 0, 0, 0, 1, 0, 0, 0, 1, 1, 0, 0, 1, 0, 0, 1, 0, 0, 1, 0]
Cruzando [1, 1, 1, 0, 1, 0, 0, 0, 1, 0, 1, 1, 0, 1, 0, 0, 0, 0, 1, 0, 0, 1, 1, 0] y [1, 1, 1, 0, 1, 0, 0, 0, 1, 0, 0, 0, 1, 1, 0, 0, 1, 0, 0, 1, 0, 0, 1, 0]
Cruzando [1, 1, 1, 0, 1, 0, 0, 0, 1, 0, 1, 0, 0, 1, 0, 0, 1, 0, 0, 1, 1, 0, 1, 0] y [1, 1, 1, 0, 1, 0, 0, 0, 1, 0, 0, 0, 1, 1, 0, 1, 0, 0, 0, 1, 0, 0, 1, 0]

4. Mutación:

--- GENERACIÓN 3 ---

1. Evaluación:


```
[1, 1, 1, 0, 1, 0, 0, 0, 1, 0, 1, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0]
[1, 1, 1, 0, 1, 0, 0, 0, 1, 0, 1, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0]
```


Cruzando [1, 1, 1, 0, 1, 0, 0, 1, 1, 0, 1, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0] y [1, 1, 1, 0, 1, 0, 0, 1, 1, 0, 1, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0]

[illegible]

```

--- GENERACIÓN 62 ---
1. Evaluación:
Bits: [1, 1, 1, 0, 1, 0, 0, 1, 0, 1, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0] | Dec: (1.99, -0.00) | Aptitud: 0.2
Bits: [1, 1, 1, 0, 1, 0, 0, 1, 0, 1, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0] | Dec: (1.99, -0.00) | Aptitud: 0.2
Bits: [1, 1, 1, 0, 1, 0, 0, 1, 0, 1, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0] | Dec: (1.99, -0.00) | Aptitud: 0.2
Bits: [1, 1, 1, 0, 1, 0, 0, 1, 0, 1, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0] | Dec: (1.99, -0.00) | Aptitud: 0.2
Bits: [1, 1, 1, 0, 1, 0, 0, 1, 0, 1, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0] | Dec: (1.99, -0.00) | Aptitud: 0.2
Bits: [1, 1, 1, 0, 1, 0, 0, 1, 0, 1, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0] | Dec: (1.99, -0.16) | Aptitud: 0.1
Bits: [1, 0, 1, 0, 1, 0, 0, 1, 0, 1, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0] | Dec: (3.13, -0.00) | Aptitud: 0.0
Bits: [1, 1, 1, 0, 1, 0, 0, 1, 0, 1, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0] | Dec: (1.99, -0.00) | Aptitud: 0.2
Bits: [1, 1, 1, 0, 1, 0, 0, 1, 0, 1, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0] | Dec: (1.99, -0.00) | Aptitud: 0.2
Bits: [1, 1, 1, 0, 1, 0, 0, 1, 0, 1, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0] | Dec: (1.99, -0.00) | Aptitud: 0.2

2. Selección:
[1, 1, 1, 0, 1, 0, 0, 1, 0, 1, 1, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0]
[1, 1, 1, 0, 1, 0, 0, 1, 0, 1, 1, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0]
[1, 1, 1, 0, 1, 0, 0, 1, 0, 1, 1, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0]

```


Cruzando [0, 1, 1, 0, 1, 0, 0, 1, 0, 1, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0] y [1, 1, 1, 0, 1, 0, 0, 1, 0, 1, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0]

--- GENERACIÓN 69 ---

2. Selección:

3. Cruza:

4. Mutación:

--- GENERACIÓN 70 ---

2. Selección:

3. Cruza:

Cruzando [1, 1, 1, 0, 1, 0, 0, 1, 0, 1, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0] y [1, 1, 1, 0, 1, 0, 0, 1, 0, 1, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0]

[illegible]

```

--- GENERACIÓN 93 ---
1. Evaluación:
Bits: [1, 1, 1, 0, 1, 0, 0, 1, 0, 1, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0] | Dec: (1.99, -0.00) | Aptitud: 0.2
Bits: [1, 1, 1, 0, 1, 0, 0, 1, 0, 1, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0] | Dec: (1.99, -0.00) | Aptitud: 0.2
Bits: [1, 1, 1, 0, 1, 0, 0, 1, 0, 1, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0] | Dec: (1.99, -0.00) | Aptitud: 0.2
Bits: [1, 1, 1, 0, 1, 0, 0, 1, 0, 1, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0] | Dec: (1.99, -0.00) | Aptitud: 0.2
Bits: [1, 1, 1, 0, 1, 0, 0, 1, 0, 1, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0] | Dec: (1.99, -0.00) | Aptitud: 0.2
Bits: [1, 1, 1, 0, 1, 0, 0, 1, 0, 1, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0] | Dec: (1.99, -0.00) | Aptitud: 0.2
Bits: [1, 1, 1, 0, 0, 0, 0, 1, 0, 1, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0] | Dec: (2.49, -0.00) | Aptitud: 0.6
Bits: [1, 1, 1, 0, 1, 0, 0, 1, 0, 1, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0] | Dec: (1.99, -0.00) | Aptitud: 0.2
Bits: [1, 1, 1, 0, 1, 0, 0, 1, 0, 1, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0] | Dec: (1.99, -0.00) | Aptitud: 0.2
Bits: [1, 1, 1, 0, 1, 0, 0, 1, 0, 1, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1] | Dec: (1.99, -0.00) | Aptitud: 0.2

2. Selección:
[1, 1, 1, 0, 1, 0, 0, 1, 0, 1, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0]
[1, 1, 1, 0, 1, 0, 0, 1, 0, 1, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0]
[1, 1, 1, 0, 1, 0, 0, 1, 0, 1, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0]

```